FACULTY OF SCIENCE, ENGINEERING AND COMPUTING

School of *Computer Science & Mathematics*

BSc DEGREE

IN

Computer Science

PROJECT REPORT

Name: Diana Marcela Poveda-Ocampo

ID Number: K1553539

Project Title: Event Team System

Project Type: Design

Date: 23/04/2018

Supervisor: Graeme Jones

Kingston University London

Plagiarism Declaration

The following declaration should be signed and dated and inserted directly after the title page of your report:

Declaration

I have read and understood the University regulations on plagiarism and I understand the meaning of the word *plagiarism*. I declare that this report is entirely my own work. Any other sources are duly acknowledged and referenced according to the requirements of the School of Computer Science and Mathematics. All verbatim citations are indicated by double quotation marks ("…"). Neither in part nor in its entirety have I made use of another student's work and pretended that it is my own. I have not asked anybody to contribute to this project in the form of code, text or drawings. I did not allow and will not allow anyone to copy my work with the intention of presenting it as their own work.

Date : 23/04/18

Signature: Diana Marcela Poveda-Ocampo

Contents

Abstract 6
Chapter 1 Introduction and Background 6
1.1.1 Background 6
1.1.2 Problem 6
1.1.3 Proposed Solution 7
1.2 Aim & Objectives: 7
1.2.1 Why is the project important? 7
1.3 Project Management Approach: 10
1.4 Discussion 10
Chapter 2 Literature Review 11
2.1 What is user experience? 11
2.2 Human centred design 11
2.3 Elements of user experience 12
2.4 Prototyping 13
2.4.1 Low fidelity prototypes 13
Chapter 3 Requirement Analysis: 14
3.1 Introduction: 14
3. 2 Actors: 14
3.2.1 Interview: 15
3.3 The System Analysis 16
3.3.1 Stakeholder Analysis: 16
3.3.2 User Stories 18
3.3.3 Use cases 20
3.3.4 Use Case Descriptions: 23
3.3.5 Functional and Non-Functional Requirements 24
3.4 Discussion: 26
4 Design Chapter: 26
4.1 Introduction 26
4.2 User Interface 27
4.2.1 Principles and Heuristic Evaluation 27
4.2.2 Page Design: 28
Low fidelity – Sketches 28
Mock ups: 29
4.2.3 Navigation and Site Maps 30
4.3 Process Modelling: 33

4.3.1 Activity diagram: 33
4.3.2 User Scenarios: 35
User Scenario: Scout 35
User Scenario: Volunteer 35
User Scenario: Event Organiser 35
4.4 Data Modelling 36
4.4.1 Entity Relationship Diagram 36
4.4.2 Data Dictionary 38
4 .5 Discussion: 41
5 Implementation: 42
5.1 Introduction: 42
5. 2 Tools, Technologies and Resources: 42
5.2.1 Sublime Text: 42
5.2.2 PhpMyAdmin: 42
5.2.3 Bootstrap: 43
5.2.4 XAMPP: 43
5.2.4 FileZilla: 43
5.2.4 StarUml: 43
5.3 Technologies: 43
5.3.1Html and CSS: 43
5.3.2 Javascript and Jquery: 44
5.3.3 PHP and SQL 44
5.3 System Architecture: 44
5.4 Implementing the database: 45
5.4.1 Creating the tables: 45
5.4.2 Populating the database: 46
5.4.3 DML Statements: INSERT, UPDATE, DELETE: 48
Joining tables together: 50
5.5 Implementation of key functions: 51
5.5.3 Login into the system: 51
..... 52
5.5.2 Add an Event 52
5.5.3 Create Volunteer Team: 53
5.5.4 Create Scout Team: 54
5.5.5 Viewing Teams: 56
5.6 Discussion 58

Chapter 6 Testing chapter: 58
6.1 Introduction: 58
6.2 Functional Tests: 58
6.2.2 Non-functional Testing: 60
6.3 Queries: 61
6.4. Usability Test 63
6.5 Console Log: 64
6.6 Discussion: 64
Chapter 7 Critical Review: 64
7.1 Introduction 64
7.2 Achievements 64
7.2.1Project areas: 64
7.2.2Persona Reflection: 65
7.3Future Work 65
7.3.1Project 65
7.4 Conclusion: 66
Appendix A: 67
Project Proposal 67
Appendix B 81
Use Case Descriptions: 81
Appendix D: 83
Create Tables 83
Bibliography 85

Abstract

Many volunteers with different types of skills are needed to run water-based activity events for sea scouts: instructors, safety boat crew, shore support, etc. Each weekend several events might be running each of which needs a team of volunteers. A system is required to store the volunteers and their skills, and to create validated teams and send out confirmation requests.

Chapter 1 Introduction and Background

1.1.1 Background

The Scouting association is one of the largest youth organisations in the UK and is the largest national scout organisation in Europe which provides challenging and fun structured activities for boys and girls. One branch of activities that the scouting association holds is the sea scouts which are water based activities for example rowing, sailing, rafting and power boating. However, to be able to do them training should happen. Through training, the sea scouts are taught how to do the activities very safely in a structured way where there are formal qualifications in which they can take. These qualifications are not actually scout qualifications but they are adopted from other organisations for example the RYA association. RYA provide a set of stage one and stage two qualifications for sailing. This allows the training to be conducted safely by completing these courses. Besides that, there is a regular weekly evening session on a Saturday for the scouts in which boating activities are held. However, if it is training activities, invitations are only made to a subset of scouts. This means if a scout has the qualifications already in a certain activity they do not need to attend the training which means that they can attend most events. Furthermore, training doesn't always consist in a classroom, practical lessons are also taught for example knowing how to paddle are one of the key elements when taking part in a water based activity.

1.1.2 Problem

Currently there is no system, as it is heavily paper-based with excel spreadsheets, this proves to be a nightmare for the event organiser. Templates of emails are sent out to volunteers, parents and scouts in which mistakes are made on them. As the predominant system currently used is paper based the problem is a difficulty in managing the spreadsheets because it isn't easily accessible for everyone to see which makes it tougher for information to be obtained and to create a team of volunteers for any event being held. Additionally, many volunteers potentially may have qualifications which meet the desired skills for the event position however all the qualifications are currently kept on one place which is makes it difficult for the event organiser to maintain and check if the qualification is up to date and keep informed of new qualifications that are being released.

Similarly storing everyone's availability is very hard to maintain because at present doodle surveys are used to ask about people's availability however there isn't any push notifications to say that a volunteer has responded to a survey. So, for this reason lots of paper and excel spreadsheets and this tool to implement are hard to be integrated because only one person can understand the workflow in managing and sending out emails. Due to the current practices, it is impossible to share the task because that one person has a workflow that he/she uses constantly and has a mechanism in processing the data which results in approaching different steps in the workflow. The lack of practice and experience for others to learn causes it very hard to teach others if they can't keep up with the same process therefore there is a need of a new system.

1.1.3 Proposed Solution

The system will include a front end which will be user friendly. This will help users to be able to navigate easily and interact flexibly. There would also be a back end to the system which will only be accessed by the event organiser who will administrate the system. This system will be more streamlined because it will allow events to be viewed by everyone and avoid misinterpreted information being stored. There are great difficulties in finding and organising volunteers to be at events therefore the system will support that through collection of availability and being able to send out invitations. Through this they will be able to report who can come and choose. Furthermore, the system will make sure that there is the right mix of qualifications and be able to see if there is the right number of acceptances so that the event organiser can create a validated volunteer team for the day of the event.

1.2 Aim & Objectives:

The aim of the project is to promote scouting opportunities for kids by developing an efficient and effective system for organising events.

The objectives:

- Designing a user-friendly website which enables users to access efficiently by April
- Create a database in which volunteer skills and qualifications are stored and populated.
- To develop a system that enables the event organiser to create schedules.
- To comply with DSDM/Atern Agile methodology

1.2.1 Why is the project important?

To go through with the project a swot analysis has been used to identify the strengths, weakness first and then the opportunities and threats that the project might face externally and internally. This technique is useful as it helps to address any issues that could affect the system at an early stage that way later in the process these limitations do not get in the way and it will save time and help to keep the project on track. However, the downside of using a SWOT is it doesn't offer any solutions on the limitations that could affect the project. In the table 1 it presents the swot analysis.

Table 1

Strengths:	Weaknesses:
• Users will be able to access and view clearly the events. • Users availability will be stored easily through a friendly user interface. • No physical storage as it will be a web application which results in low cost. • Event invitations will be easier to manage. • The system will be able to restrict views depending on the user type.	• Lack of technical knowledge may affect speed of completion. • Web hosting company might get hacked. • Back up storage can get tainted which can result in losing data • Restricted time to complete the system • Other projects involvements may cause delay
Opportunities	Threats
• The system can be expanded for the use of other sport events other than water based events • Promote scouting opportunities for kids. • Increase the number of participants in events and the club becomes more recognised/accredited for a well organised event.	• Revolution in technology which results in the system in becoming out dated • Risk of hackers stealing information about participants.

To be able to turn the weaknesses into strengths it would be helpful to look up and go through some PHP tutorials online to be able to increase my knowledge that way it wouldn't affect the speed of the completion of the project. To be more organised, it would also be helpful to create a deliverable schedule that way the project can be delivered on time and other projects wouldn't get in the way. In addition, to be able to overcome web hosting companies getting hacked it would be helpful to begin with working in the local host that way changes being made could be used easily on the local server. Overall looking at table 1 the list of points made about the strengths of the projects internally outweigh the weaknesses, which means it is safe to say that the project can continue.

Once a SWOT analysis was carried out a risk analysis was later produced to be able to analyse the potential risks that the system may face. Most of the risks were devised from the swot analysis that was done at the start of the project as it is important to address the

dangers that would affect the system. A contingency plan was also produced on each risk as devised recommended plans to be able to avoid the risk if it happens. All risks were measured against a quantitative measurement, where it outlines the likelihood of the risk happening and the impact it will do to the system if it occurred. All the risks which have a low likelihood and impact do not do anything noteworthy to the project, however they may take up time in the long run which could result to a cost to the project. On the other hand, all the risks which have a medium likelihood and a high impact could have a negative impact to the project if it isn't properly considered. But since all the risks don't have a high likelihood and all the other risks identified were given a contingency plan then it is considered safe to proceed with the project.

Risk	**Likelihood**	**Impact**	**Mitigating**	**Contingency**
Not getting support from my supervisor.	low	high	Have regular meetings with the supervisor to discuss ideas.	Use other resources example books or internet. Seek help from other supervisors
Poor communication with my client	medium	high	Need to contact with my supervisor to discuss about my project immediately and find alternative ways to contact my client eg skype	Make sure to contact another person within the organisation to be the client.
Losing important documents	medium	high	Need to make sure that regular backup is done weekly and many copies are saved in different places.	Recover from back-up
Lack of technical skill	medium	medium	keep utilising resources online or books.	Seek help from lectures who have the required skills
Database failing	low	High	Hosting company back-up	Recover from back-up

1.3 Project Management Approach:

The methodology used to manage the project and the software is DSDM/Atern Agile Methodology. The DSDM framework was used to manage the project, the reason for this is because it consists of three sequential phases which are; the pre- project, project life cycle and post project phases. In the pre- project phase, the problem to the project is identified and any issues that would affect the system are identified also too. In the project life phase, it consists of making iterative changes throughout the process which means the design stage and implementation change and analysis stages can be done iteratively and finally in the post project phase it ensures that the system is operating effectively and efficiently. (what is DSDM? , 2018)

The goal for using DSDM/Atern is to deliver projects on time and on budget while adjusting for changing requirements along the way. Since there is a fixed scheduled time for the project report submission DSDM/Atern enables to provide an active user involvement always which means that client meetings will be done every week ensuring that the development of the project can be produced iteratively. By having user involvement, it provides sufficient feedback to the developer which encourages using one of the core techniques of DSDM such as timeboxes. Using timeboxes it will enable to focus on what is really needed to meet the deadline therefor a deliverable schedule was created where every week a deadline was set to be met. Another technique that is beneficial to use in the project is prototyping which will be used in the design stage. The use of prototypes will help to provide feedback from the end users and ensure that product will provide a good user experience. Moreover, another technique that would be used is MoSCOW methods which is ensures that the requirements elicited are prioritised.

A methodology that was disregarded was the waterfall methodology because it does not go through an iterative process which means that, at any stage if changes were to be made it couldn't because once a stage is completed, changes cannot be altered. This means that if requirement changes were to be changed later in the process it would not be allowed which would result a high impact to the project. As this methodology is not iterative the client does not give sufficient feedback to the developer which may result the client not being pleased with the final product.

A software methodology that was rejected is extreme programming because it uses the concept of 'pair programming' and this project is not paired. Furthermore, it uses a lot of daily user involvement which the client and developer would find it hard to manage due to other project deadlines that need to be manage too. Furthermore, by using extreme programming in the design stage it uses a 'simple design' which means no design techniques are produced this would affect the system since it will miss out on the way the users will experience and behave using the product which could evidently lead to a lot of changes in the process.

1.4 Discussion

During this phase of the project several client meetings were arranged to be able to understand the problem to the project. It is important to be able to outline the aims and

objective of the project, that way there would be a specific goal that would be realistic, measurable and achievable to accomplish within the required time. A swot analysis was carried out which concluded the strengths outweighed the weaknesses. One of the potential threats included overcoming web hosting companies from getting hacked, which as mentioned to prevent this from happening the project will be working in the local host that way changes being made could be used easily on the local server. A risk analysis was then carried out after the swot analysis as it enables to outlines any dangers that occur towards the project. Every risk that was identified had a contingency plan therefore it was safe to say that the project could continue. In addition, it was important to identify what methodology was going to be used to manage the project. An agile approach was used because it allows iterative changes throughout the process cycle, whereas using the waterfall methodology once a stage is completed there is no turning back to make any changes.

Chapter 2 Literature Review

2.1 What is user experience?

Due to the scope and versatile nature of user experience it has led to numerous definitions and viewpoints. The International Organization for Standardization (OSI) defines user experience similar as a 'person's perceptions and responses resulting from the use and/or anticipated use of a product, system or service' [Author, 1997]. However, Hassenzah and Tractinsky suggest that user experience covers 'products, systems, services, and objects that a person interacts with through a user interface' and that face to face interaction between humans is outside the scope of user experience [Hassenzah and Tractinsky, 1987]. User experience is vital especially with web applications as it is important for the user to be able to navigate easily and can understand how to use it. By neglecting the user experience, it makes it harder for the user to achieve their user goals

2.2 Human centred design

Before designing an interactive product, there needs to be an understanding of what is required to be able to support the user's interaction this means requirements need to be elicited. By underlying a good interaction design there needs to be the user's involvement and feedback at all stages of the design process which results in gathering the user requirements, testing and evaluating the final products, systems, services that the user interacts with. The advantage of using a user centred design approach is that it will guarantee that the product will be suitable for its intended purpose in the environment in which it will be used. This will help to provide better support to the product as it would lead to greater effectiveness, efficiency and increased safety. Furthermore, by having the users involved they would know what to expect at an early stage and that their suggestions have been considered. This leads to a sense of ownership for the final product that often results in higher customer satisfaction and smoother integration of the product into the environment. [Preece Rogers,2000]

There are many user centred design models such as goal directed design, contextual design and the elements of user experience. All these models agree to the ISO (2010) framework for user centred design and its principles;

- the users are involved throughout the whole design process- this requires the user to be throughout the design and development process.
- the design process is iterative – this is when problems are found in the usability test they are fixed and then more tests and observations are carried out to see the effects of the fixes.
- the design is based on the understanding the users, tasks and environments- this is important to understand who the users will be by observing how the users will be able to perform their normal tasks in their own comfortable environment.
- The design is refined by user centred evaluation- this requires measuring the utility of the system by using usability metrics which includes measuring the user's satisfaction and performance.

2.3 Elements of user experience

Garret presents a conceptual model for a user experience design, breaking it down into five different planes; strategy, scope, structure, skeleton and surface. These planes provide a conceptual framework for talking about user experience problems and the tools we use to solve them. Each plane depends on each other starting from the bottom to the top. As you go up the plane the issues become more concrete and less abstract, for example on the strategy plane we are concerned whether the user needs are the goals. Garret states that it is important to have an effective communication as a key factor when building a website as there is no point having a website that does not function properly and users not being able to figure out to make it work. Therefore, by creating an efficient user experience a user centred design approach should be used this would allow the user to be involved throughout the whole the design process and be able to understand more about the user's expectations.

Garret came up with five elements of user experience for websites due to its basic duality that websites be functional application which meant on focusing how users think about completing the tasks or information medium which is concerned about what information is offered by the site. For this reason, the site is split into two where the left side of the model are elements specific to the functional products and on the right, are elements which are specific to the information products.

On the strategy plane user needs is important to determine the product strategy as it is important to understand who the product is being built for and why they are going to use it, alongside with the product objectives which can be business goals for example an increase in turnover for a site. The scope plane describes the functional specifications of the product which specifies what the system should do and the content requirements which are the information which is needed to be included in the content that is being developed. The

structure plane explains the interaction design which defines how the user interacts with the product and how the system behaves in response to the user interactions and the information architecture which explains the arrangement of the content elements and how they are organised. By having a good interaction design it helps the user accomplish their goals. The skeleton plane is split into three components; interface design, navigation design, and information design these components enables users to interact and navigate with the system. The surface plane considers sensory design which considers on how the product will look.

2.4 Prototyping

Having established the requirements, the next step is to establish design ideas through prototyping and evaluation cycles. To be able to produce an interactive design there needs to be a way where the users can evaluate the design this can be achieved through prototyping. 'The activity of building prototypes encourages reflection in design' as described by [Schon, 1983] which means it is an effective way for designers to explore design ideas amongst team members. There are different prototyping techniques that a designer can use some of which include;

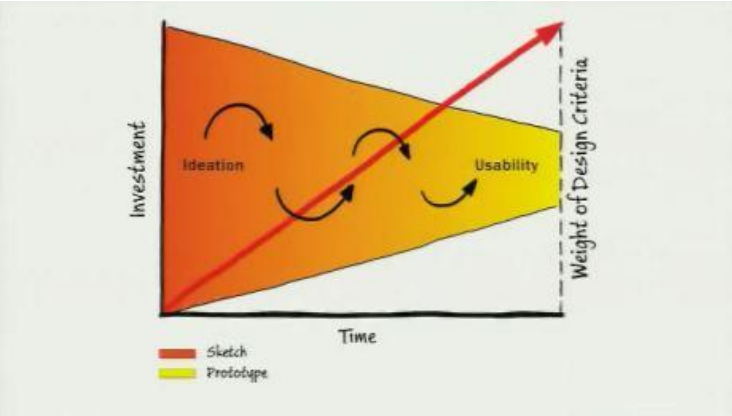
- Low fidelity prototypes
- High fidelity prototypes
- Mixed fidelity
- Multi fidelity

2.4.1 Low fidelity prototypes

Low fidelity prototypes are prototypes which occur at an early stage of the design process. These are useful because they are quick and cheap to produce as they are hand drawn. According to McUrdy "Low-fidelity methods have received a great deal of recognition in the field for their ability to validate designs and predict large problems at an extremely low cost" [Mcurdy 2006]. This shows that it is important to be able to sketch prototypes as it enables early visualization of alternative design solutions and helps to provoke improvement through iterations as designers are more comfortable at suggesting changes. However, some may argue that many people may find it difficult to engage in sketching as they are inhibited about the quality of the drawing but as Greenberg states "Sketches do not have to be pretty, beautiful, or even immediately understandable by others. However, you should be able to explain your sketches and ideas when anyone asks about them" [Greenberg,2001].

However, Bill Buxton disagrees that sketches are not prototypes and should be done at the early stages of the design process, whereas prototypes are more concentrated at a later stage. Figure 1 below presents Bill Buxton '*design funnel model*' in which he emphasises how sketching is important and should be done at an early stage of design and the investment into prototyping is higher than the investment in sketching. At the start of the funnel, sketching dominates the process, since there are still a few concepts to explore as

things are still quite uncertain about the way the design should be. Furthermore, the circular arrows reinforce that the whole design phase is an iterative, user-centred process.



2.5 Conclusion:

To conclude this literature review has given the insight of how it is important to think about the user experience when using a system. The use of paper prototypes is important in a design process as it enables to emphasise how the flow of the web pages should be and distracts any technological functionality to the prototype. Therefore, taking into consideration on Mcurdy' view on low fidelity, the use of low fidelity prototypes would be used in the design process as a part of understanding how the user interface design should be before actually moving into a high-fidelity prototype.

Chapter 3 Requirement Analysis:

3.1 Introduction:

This chapter will focus on eliciting the requirements for this project, this will be done using various analysis methods to determine the requirements of the system. It is important to identify both functional and non-functional requirements to comprehend what is needed for the project so that it can be fully accomplished. The techniques carried out in this chapter include; a stakeholder analysis, use case diagrams, user stories, interviews, use case texts. Furthermore, by applying these techniques, it will enable to produce a list of requirements where it would be prioritised using the MoSCoW method which would help to identify which requirements should be prioritised in the system.

3. 2 Actors:

To be able to identify the actors that are involved in the system an interview was carried out with the client this would ensure and be aware that any changes made throughout the system these actors identified will be affected.

3.2.1 Interview:

1) **Who is involved in the events?**

There are scouts, these are the children which take part in the events, the parent takes the children to the events. There are also volunteers, these are people who can happen to be available to help on the day of the event and are willing to help because of their skills and lastly the event organiser, who organise and manages the events.

2) **What do you mean by volunteers having skills?**

Volunteers are usually instructors who have skills which they teach such as kayaking, sailing, rowing, diving. There are leaders who are also instructors too who teach for example powerboating they have boat safety qualifications. So basically, volunteers play roles and those volunteers have a set of skills.

3) **Are all parent's volunteers?**

Nope, there are parents who don't take part. Every scout has a parent and every parent have a couple of scouts who attend. However, some of the parents are volunteers but not all parents are volunteers. For example, in three weeks' time I won't have a scout anymore because he is going to university.

4) **So, are you a volunteer and what is your role?**

Yes, I am and I'm an instructor for sailing.

5) **What is an event?**

An event could be a sailing practice that happens on a day at a specific place and it needs people to be there. Therefore, there needs to be a relationship between events and users.

6) **How is the event organiser going to use the system?**

We might decide that in a term we might have all the set of activities planned. We gather the dates, we will appoint some organisers. So, one of the event role will be the organiser who will have nothing to do on the day of the event just organise the event. The event organiser will put out invitations to scouts. Parents and scouts need to go back to the website and say yeah! -That they will go to the event. So, if there is a stage 2 sailing happening the event organiser will need make sure not to invite scouts who already have the qualification.

It is clear to understand that the following actors which are involved in the system are;

- The Scouts – the scouts are who take part the day of the events.
- The Event organiser – the event organiser oversees organising the events on the day.
- Volunteers- the volunteers help on the day of the event. They are assigned to roles on the day event according to their set of skills they have.

3.3 The System Analysis

3.3.1 Stakeholder Analysis:

The stakeholders are the actors who are involved in the process of the system but they can be either internal or external too. To be able to identify the key stakeholders which are important to the system, a power interest grid would be used as a technique to be able to find out what exactly do the stakeholders expect from the system and what are their benefits from it.

Since there are several stakeholders each will have an influence, power and interest to the project. To be able to identify which are the important stakeholders their interest would be high and their power, vice versa if their interest is low and their power is low these stakeholders would be categorised as the least important stakeholders to the system. The stakeholders in this project are:

Volunteer:

The volunteer is the person who is willing to help on the day of the event. Each volunteer will be part of a group of volunteers that will form a team according to their confirmed availability. As the club consists of volunteers the client will be represented by Graeme Jones. The client will be attending weekly meetings to discourse about the project and receive feedback.

The volunteer will be able to receive an invitation to an event efficiently as there would be quicker response time in confirmation. They would also be able to view the event in a responsive format which will benefit from a better user engagement rather than paper based.

Event organiser:

The event organiser is the person in charge of organising the event and making sure that on the day of the event there is set number of volunteers which are available to attend and help on the day.

The event organiser will be able to create an event and send invitations out to the scouts and parents this will minimise the problem with the current paper based system because presently the event organiser sends out too many errors on the templates of emails. The event organiser will also be able create a team of volunteers.

Scout:

The scouts will be training up for upcoming events. Therefore, they will be on the lookout for any events that are being held at the club so that they can participate and compete. They can be used to test the website as they will have their own dashboard and provide feedback.

The scouts will be able to view the events through their own dashboard this will benefit them as they will be able to confirm their attendance for an event rather than responding to

many templates of emails which is one of current problems with the paper based system. As each scout has a certain level of qualification they will be able to update their qualification status and apply for a specific event.

Parent:

The parents approve their scouts' availability for the day of the event. The parents are also volunteers which means that they can be part of the team of volunteers to help on the day of the event, however they do not require a personal qualification to do this.

The parents will be able to view the event and confirm attendance for that specific event through their own dashboard this will be more user friendly for parents as they wouldn't have to navigate around the sites for events. This will minimise their response times in confirming the events which will aid the event organiser in being able to create the event smoothly and improve organisation. This will help tackle one of the problems that is happening with the paper based system.

Sponsors:

The sponsors are an external stakeholder which are committed to coming to watch the events and provide support to the scouts that take part. They enable to create more awareness about the type of sports that are carried out on the day of the events and help to provide full support so that challenging activities can be organised and unique experiences to 6 to 25-year olds.

The sponsors will be able to view the number of scouts and volunteers that are registered in the system which will help them give an idea of how much the organisation has overcome over the past years. Through this they can benefit to create cultural awareness in the organisation and provide excellent support for both the scouts and volunteers on the day of the events.

Figure 1 below shows the power and interest of each stakeholder in the project. By observing the figure above the event organiser is for stakeholders who will have a high interest and high power in creating the events. Therefore, this means that they will hold the most impact on the project success, hence their expectations should be managed closely and engage on a regular basis. The scout stakeholders will be put in the high interest and low power quadrant because they would be monitored and kept informed when events will be running. The volunteer stakeholders will have a high interest and some level of influence because they would be the ones who will be influencing the event organiser in building a team to help on the day of the event therefore it is important to keep them satisfied. The parent stakeholders will too have high interest and some level of influence as they would need to be able to approve their availability as they would be the ones taking their scout to the event. In addition, as mentioned before there are also the external stakeholders which are the sponsors their interest would be somewhat low as they wouldn't be interacting with the system however they would potentially have high power because if there isn't enough support for the events to happen such as the facilities and equipment's and mainly support

for the volunteers then there really wouldn't be an event organised as the volunteers are who support Scouts in a wide range of roles.

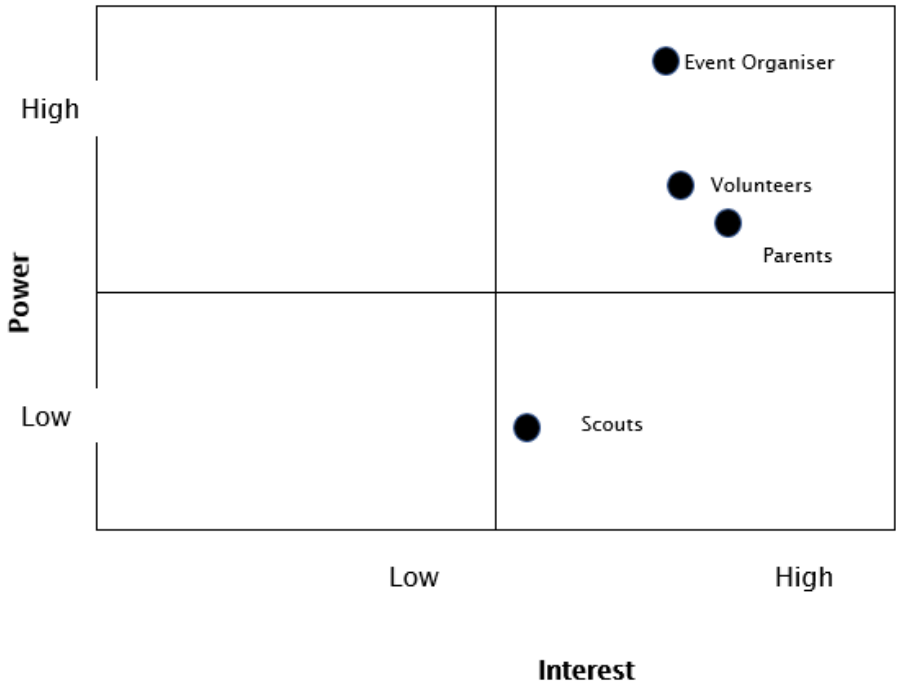


Figure 1

3.3.2 User Stories

In summary of the stakeholder analysis that was produced it was concluded that the key users of the system will be; volunteers, scouts, and event organisers. As the software approach to this project is an agile approach a certain technique commonly used in agile approaches was used such as a user story. A user story helps to describe the type of user, what they want from the system and why they want it. The reason why this technique was chosen was because it conveys a simple language format through "As a [role], I want [feature] because [reason]. " (Emily Grace Adeshiah,2015)

This format is very clear and easily understandable for the client and to the users as it moves away from writing about the requirements to just talking about them in this format. Therefore, for each identified stakeholder a user story was created.

User story:

As a/an	**I want to**	**So that**
Scout	view events	I can have a good understanding of what events are available to

		participate in.
Scout	Register event	I can be able to participate my interest in the event.
Scout	Log in	I can be able to view and register for an event

User Story:

As a/an	**I want to**	**So that**
Volunteer	View event	I can have a good understanding of what roles are needed for an event.
Volunteer	Add availability	The event organiser knows I am free on a specific day
Volunteer	Add qualifications	I can acknowledge my skills that I have.
volunteer	Confirm attendance	I can sure that I can support on the day of the event
Volunteer	Edit availability	I can update and ensure that I am invited to the correct events

As a/an	**I want to**	**So that**
Event organiser	Add events	I can provide training activities for boys and girls,
Event organiser	Invite scouts	I can create an event for scouts to attend too.
Event organiser	Invite volunteers	I can have a general over view of how many volunteers are available for an event.
Event Organiser	View availability	I can invite volunteers and scouts to an event
Event Organiser	Add roles	I can assign a volunteer with a role on the day of the event
Event Organiser	View qualifications	I know who is qualified to play a specific role at the event.
Event organiser	Delete event	If there aren't enough volunteers to help on the day of the event then the event gets removed
Event organiser	Edit event	I can update the event description.

User Story:

3.3.3 Use cases

To be able to understand more about the functionality of the system, use case diagrams were created which composed of separating the actors, the system and the use cases this is because it helps to observe easier the interaction between the user and the system. Each use case is drawn as an eclipse which describes a set of actions made by the actors. The benefit of using the use cases is that it can help to break down the requirements into shorter scenarios just like the user stories, however each use case is described in a verb noun format.

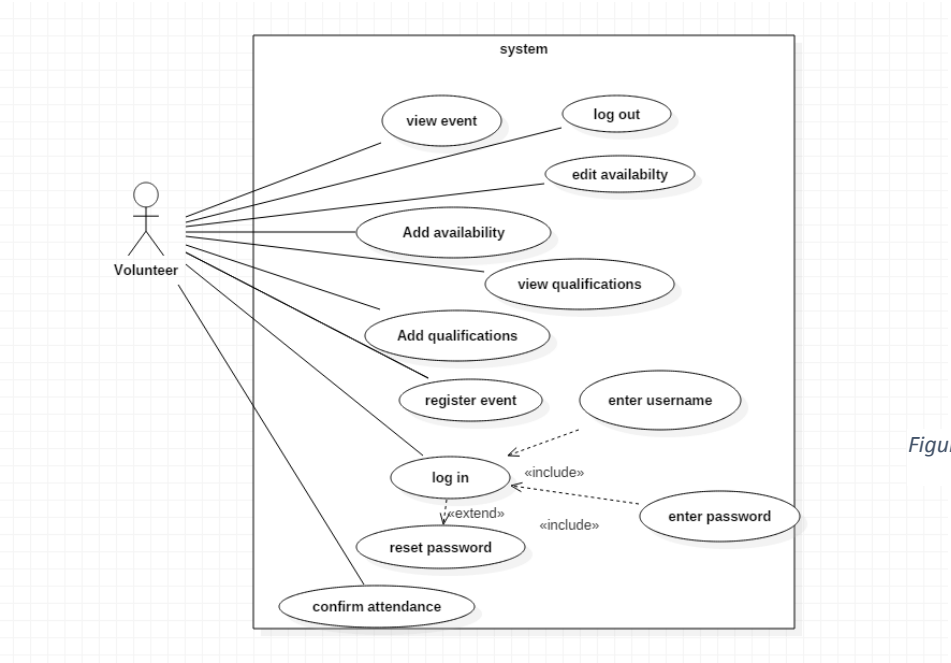


Figure 2

In figure 2 it displays the interaction the system has with the volunteer. The volunteers can view the events once they have logged into their account. To help on the day of the event, each volunteer must have a required skill to be able to be assigned a specific role. Therefore, each volunteer can manage their skills, which discussed in previous interviews that the term 'skill' are the set of qualifications that each volunteer has. The term 'manage' covers the add, edit and delete functionality that each volunteer can perform with qualifications. For the volunteers to be able to attend the events they would need to register first that way the event organiser knows that they are interested and it also assumes that they are available on that day unless they state otherwise.

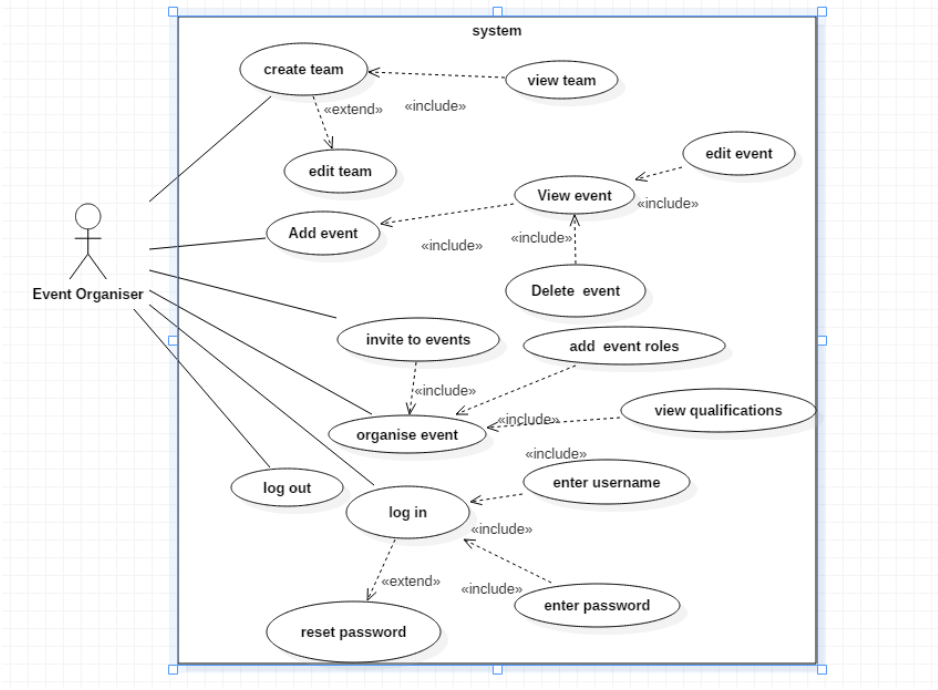


Figure 3

In figure 3 it displays the interaction that the event organiser has with the system. The event organiser will be able to manage each event by; adding, editing and deleting them. However, to be able to organise the event the event organiser would need to send out invitation emails so that the volunteers and the scouts are acknowledged that an event has been created. Once an email has been received both users would be able to register for the event which acknowledges their interest. The event organiser will also be able to create teams for each event, where they would be able to view the team and add potential participants to the team. Furthermore, the event organiser would need to create volunteer teams which would help on the day of the event, to be able to do this the event organiser would need view their qualifications and through this they would be able to decide what sort of role they should be assigned to. In addition, as the event organiser is one of the key stakeholders they must have their own account to be able to organise and manage the events.

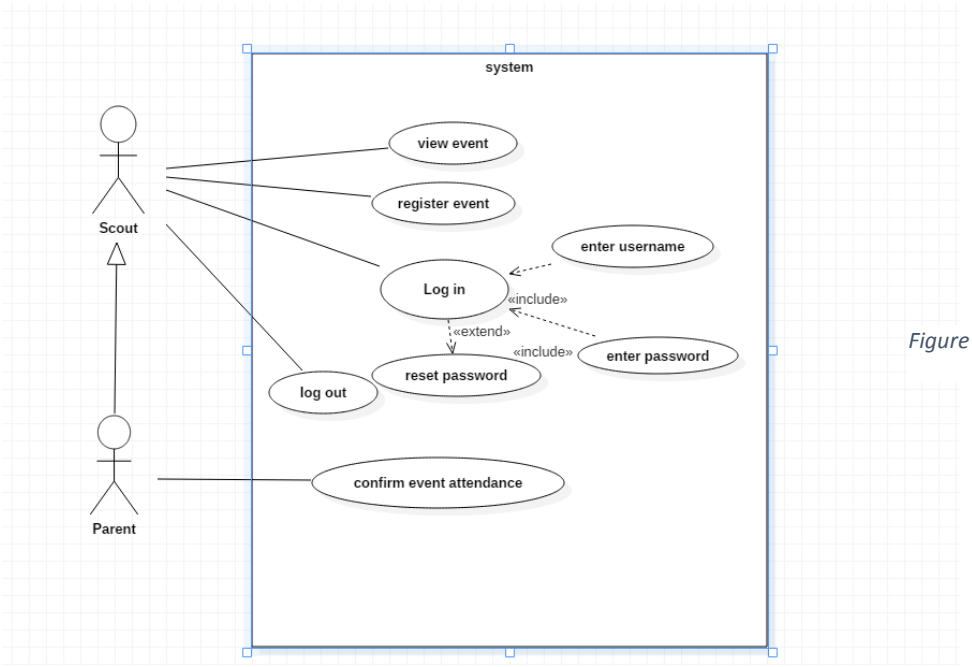


Figure 4

In figure 4 it displays the interaction that the scout and the parent has with the system. The generalization arrow is used to indicate that the parent inherits the same roles and can share the same goals as the scout. The scout would be able to log in into the system and view the events in which they are able to register their interest for the event. The parents take the scouts to the events therefore on their behalf they would confirm their attendance on whether the scout will still be able to take part on the day of the event.

3.3.4 Use Case Descriptions:

Use case descriptions were created alongside for each use case that was identified in the use case diagrams. Each use case descriptions help to consider any alternative flows that the user may have when performing the set of goal. It begins with defining the name of the actor and any pre- conditions that need to happen before performing the use case. Furthermore, it outlines any steps that would occur if an error occurs on the basic flow which helps the user be able to get back to where they left off in the flow. In table 1 it shows an example of a use case description for the event organiser with a written description on how they would perform adding an event on the website.

Table 1

Use Case ID:	1
Use Case:	Add event
Description:	Actor wants to add a new event
Actors:	Event organiser

Pre-Conditions	The actor should be logged in
Post Conditions	An event description is added
Basic Flow	1. Once the actor has logged in successfully it goes to its dashboard and presses the 'create event' tab. 2. It leads to a screen where the actor fills in a form which provides details of the event. 3. Selects 'add event' button and displays a success message
Extensions	1a. The actor has entered the wrong username or password which doesn't allow the actor to enter the system 3a. The actor will get a validation message if the event form is not filled in properly.
Performance	Adding the event should be created within 5 seconds
Priority	Must
Trigger	none

3.3.5 Functional and Non-Functional Requirements

The list of requirements was split up into functional and non-functional requirements. The functional requirements were collected through the use cases. To be able to identify which requirements should be prioritised for the project a MOSCOW method was used. The reason this technique was used was because since this project has a fixed deadline time it is vital to understand the relative importance of the requirements as it keeps the progress so that deadlines are met. The requirements which are 'must' are requirements which are essential to the system so that they can be delivered. For example, being able to add an event in the system is vital because scouts need to be able to attend to events and if no events are created then there is no use in creating teams for both scouts and volunteers. The requirements which are 'should' are requirements which are important to the system but not vital. An example is being able to edit an event, the event organiser should be able to change certain attributes on the events if changes need to be made drastically. The requirements prioritized with 'could' are requirements which leave a less impact than the 'should' requirements and the 'won't have' are requirements which will not be delivered in the project. In table 2 the list of functional requirements is displayed.

Table 2

<u>Functional Requirements</u>

ID	Requirement	Prioritisation	User Group
1	Add event	Must	Event Organiser

2	Edit event	Should	Event Organiser
3	View event	Must	Event Organiser, Volunteer, Scout
4	Delete event	Should	Event Organiser
5	Add event role	Must	Event Organiser
6	Invites to event	Must	Event Organiser
7	Log in	Should	Event Organiser, Volunteer, Scout, Parent
8	Log out	Should	Event Organiser, Volunteer, Scout, Parent
9	Register event	Must	Volunteer, Scout
10	Confirm event attendance	Must	Volunteer, Parent
11	Add availability	Must	Volunteer
12	Edit availability	Should	Volunteer
13	Add qualifications	Must	Volunteer
14	View qualifications	Must	Volunteer, Event Organiser

In table 3 it shows a list of non-functional requirements where it describes on how the system works. They were created by categorizing them into different non-functional groups such as; performance, usability, reliability, supportability this helped to suitably think about what sort of requirements would help the users experience when interacting with the system.

Non -Functional Requirements

Requirement	Prioritization
The login password must consist of upper case letters and lower-case letters of more than 5 characters	Must
The system must have a consistent user interface design in all pages,	Must

Response time for logging into the system must be less than 5 seconds.	Must
The system must be available to all the users at any time.	Must
The system shall be supported by many browsers.	Should
The system could register users interest for an event before sending out invitations.	Could
The system must support at least 100 users using the system simultaneously.	Must
The system should provide easy access to the event information.	Should
The database must put up with at least 100 records of data being inserted.	Must
The system should ensure that the correct information displayed is visible according to the right user type.	Should
A physical backup of the database should be performed at least once a week.	Should
The system must include error messages or validation checks to ensure the user is completing a task properly	Must

3.4 Discussion:

During this stage functional requirements and non-functional requirement to be able to build a system that matches the end user's needs. Moreover, since the methodology chosen conveys an agile approach some requirements may change throughout the project which may be a problem if changes are made too late within the process of the project. Therefore, to ensure that the right requirements were elicited an interview was made with the client to be able to understand the client's needs in more detail. A stakeholder analysis was also created to be able to identify the key stakeholders which was concluded that the scouts, event organiser and volunteers will be using the system more since they have a higher interest. Since an agile methodology was chosen to manage the software side of the project user stories were used to be able to demonstrate the type of users that will be using the system followed by what they want from the system and why they want it. This technique enabled to then create a use case diagram to demonstrate how the user will interact with the system through use case.

4 Design Chapter:

4.1 Introduction

This chapter will outline the design of the user interface of the website and the database design which was decided after gathering the full list of requirements in the requirement

chapter. The design of the user interface of the website will consist of several different webpages that would show the key information that the user would need and show on how it went from several sketches to a visual representation and the database design would discuss on how it went from a conceptual diagram to an ERD diagram. Furthermore, sequence diagrams would be used to be able to depict the objects and classes that are involved in a use case and activity diagrams would be used to help the client to understand the series of actions that the client will perform when using the system. By designing both the user interface and database design it will give a clear idea to the client on how the system should look and what should be stored.

4.2 User Interface

4.2.1 Principles and Heuristic Evaluation

Design decisions were made when designing the webpages these decisions were made by the client and suggestions were made which were collected from Jakob Nielsen 10 general principles for interaction design. To be able to design the layout of the user interfaces of each web page, a structured layout was used so that the pages looked consistent. Therefore, Jakob's Nielsen's 10 general principles for a user interface design was used to evaluate the user interface; (Nielsen, 2010)

- Visibility of system status
- Match between system and the real world
- User control and freedom
- Consistency and standards
- Error prevention
- Recognition rather than recall
- Flexibility and efficiency of use
- Aesthetic and minimalistic design
- Help users recognize, diagnose and recover from errors
- Help documentation

Having looked at the heuristic evaluation on the webpages it was summarised that the layout of the pages had to look consistent that way it provides a better user experience to the user as they would be familiar on the pages and easier to perform the actions. Furthermore, it should include a minimalistic design which means that it should not contain any information that is irrelevant to the user to see and if there are errors for example when filling out forms the data inputted should be validated and correctly displayed to the user.

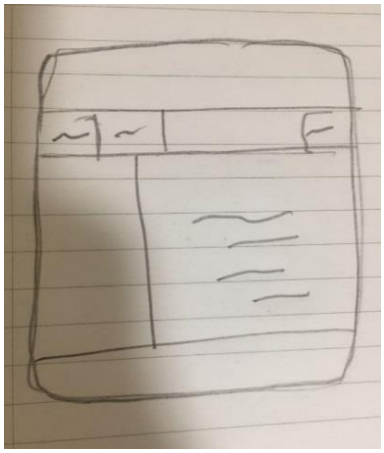


Figure 4.1

In figure 4.1 each webpage is going to consist with this layout throughout the whole system. By applying Nielsen's principles, the layout included a header which includes the

name of the system, for example on the homepage. Below the header there would a horizontal navigation bar which would include the names of the links that each page would have which would depend on the user's rights. The reason why a horizontal navigation bar was used is because in 2009 Nielson Normal Group conducted a research into how people read and according to the research it revealed that people tend to read web content in an "F shaped Pattern" therefore this choice was made. (Nielsen Jacob, 2010)

The centre of the page would include the main content section that would be displayed according to the page and a left side navigation bar this would include the functions that user can perform once on the page. The reason why there is a left navigation as it is quicker for users to scan which leads to being more efficient. There would also be a footer which would contain the copyright information of the system and any links and a contact information if needed and links if the user can't find what they want in the page.

4.2.2 Page Design:

Low fidelity – Sketches

Sketches were used because it is easy and fast to draw layouts and help to find the best ideas and solutions to a design problem. Several sketches were created to show what key pages would be needed in the system and how the page would look and allow an initial layout of the pages. Furthermore, using paper prototypes at the start of a design process it also helps to portray the workflow rather than focusing too much on the visual design such as any images that the system should include as this is not really an important aspect to the system. However, by using paper prototypes it is a very useful way in gathering feedback by the users and make any iterative changes very quickly.

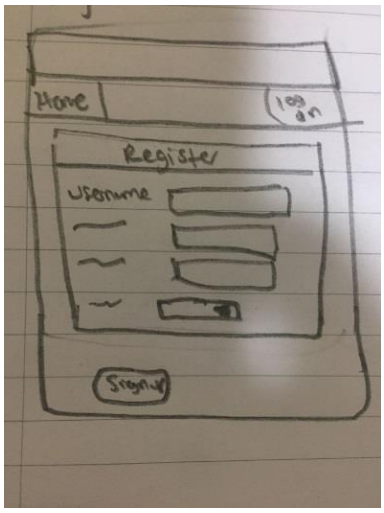
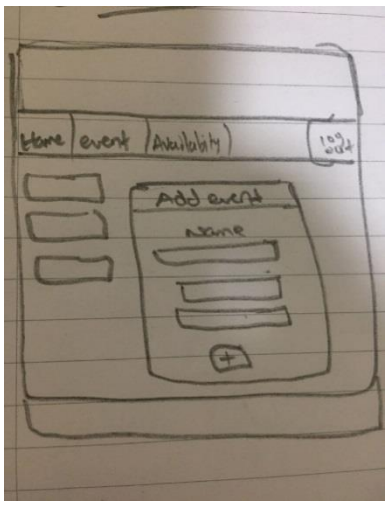


Figure 4.2

In figure 4.2 it presents the registration page which would allow each user to input their details and select what user type they are as each user type would have restricted views on their dashboard. There would also be a validation check on whether all fields are entered and selected and if their username has already been taken before. This would ensure that valid details have been filled out properly and the user can then select the submit button and be redirected to the login page.



In figure 4.3 the event organiser will be able to create an event by filling in the event form which would contain the name of the event, description, date and time and location of the event, once submitted the user will be notified through an alert message at the top of the page for validation. Depending on the user type the events page will have different functions displayed on the side navigation bar. For example, the scout wouldn't be able to add events however it will have the ability to view the events and register for the events, whereas for the event organiser they would need to be able to manage the events, view and create a team of participants and volunteers.

Figure 4.3

Mock ups:

After producing several paper prototypes, medium to high fidelity wireframes were created to represent a visual content of the functionalities in a static manner. These wireframes included visual details such as colour and design to give the user a more realistic design of how the system would look like, which this was beneficial as it helped my client to review the visual side of the system.

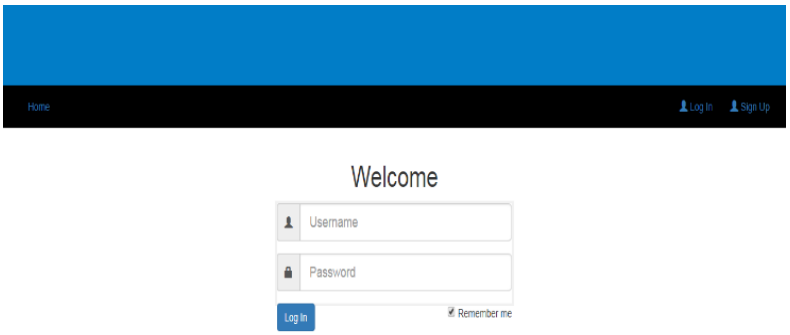


Figure 4.4

[redacted]

Figure 4.4 shows the login page, it differs with all other pages because it only has two links on the navigation bar this is because at this point of the page the user has no access rights. Therefore, once they have logged in they will be able to see what access rights they have, this depends on the user type. If the user types in the wrong credentials then the system will notify the user with a red error message. By doing this it would allow the user to recognise that they have made an error, this applies with one of Nielsen's principles as he states that the system should 'help users recognize, diagnose and recover from errors'.

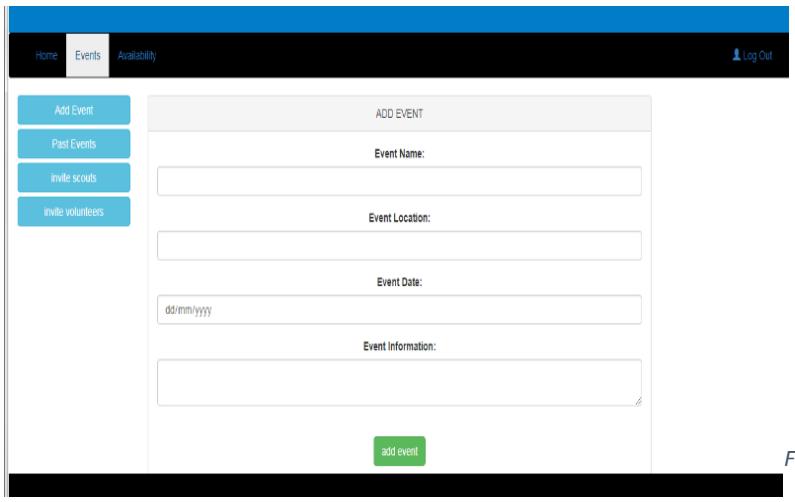


Figure 4.5

Another wireframe that was produced based on the sketches that was created was adding an event. In figure 4.5 shows how the user interface would look once the system is implemented. The colour scheme was kept consistent for all pages, which consisted a blue header, a black navigation bar and a white background on the main contents page.

The reason the header was blue is because at the system is for water based events so it was decided to keep that colour to represent the water. The navigation bar was kept black as the colour contrasts with the white main content background, this is helpful as the when hovering over the links on the navigation bar with a different colour it helps to direct the viewer exactly where would want to go. The main content was left as a white background was because it is a neutral colour and Nielsen's mentions that an optimal legibility requires black text on a white background which is called positive text. (UXPlanet, 2012)

4.2.3 Navigation and Site Maps

Sitemaps are very helpful tools which helps to provide a navigation of the website by helping to allow the users to see all available content areas on one page and gives them an instant access to the site pages. The content on the site map comes from the functional requirements that was elicited in the analysis stage. In figure 4.5 the site map begins with the users being able to log in, which means that each user should be able to register themselves and attain a username and password to use the system. In the second layer, are the name of the user types that would be accessing the system. The third layer consists the name of the menus which would be displayed on the navigation bar, since there are different users each user will not have the same navigation links as each user has a different profile and are restricted to certain functionalities.

In figure 4.6 it shows an example of how the event organiser can create an event. The event organiser can click on the event navigation button which will give the user an option to create an event, this consists of being able to manage the event by; editing, view and deleting the event. Once the event is created the event organiser will be able to invite scouts to participate in the event and volunteers to form a team. After invitations are sent the event organiser can view and select the scouts and volunteers to be in the event.

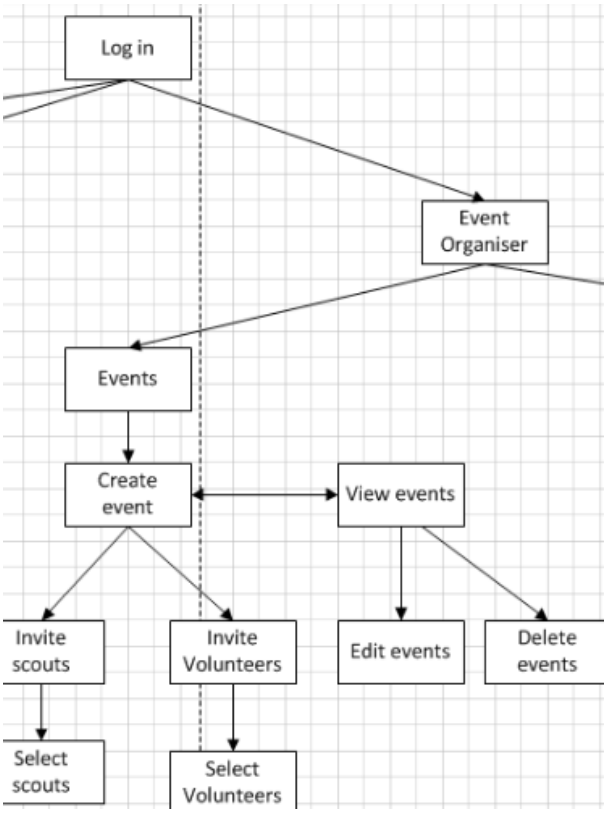

Log in
Event Organiser
Events
Create event
View events
Invite scouts
Invite Volunteers
Edit events
Delete events
Select scouts
Select Volunteers

Figure 4.5

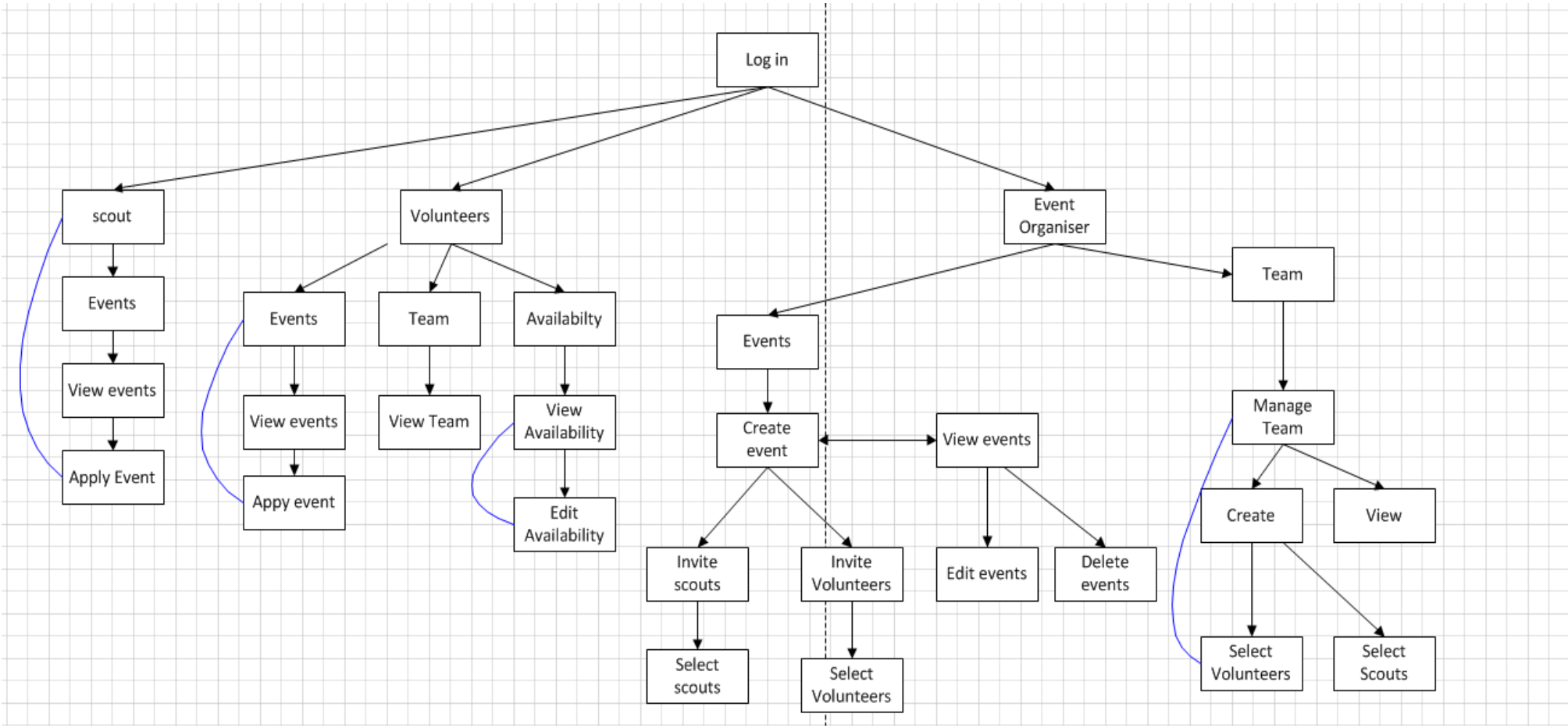

Log in
scout
Volunteers
Event Organiser
Events
View events
Apply Event
Events
Team
Availabilty
View events
Appy event
View Team
View Availability
Edit Availability
Team
Events
Create event
View events
Manage Team
Create
View
Invite scouts
Invite Volunteers
Edit events
Delete events
Select scouts
Select Volunteers
Select Volunteers
Select Scouts

Figure 4.6

4.3 Process Modelling:

4.3.1 Activity diagram:

Activity diagrams are used to describe the flow of the system. The activity diagrams were chosen from the use cases that were derived in the analysis. Each diagram is composed of showing the steps and any alternative steps that may occur in the flow. Therefore, various activity diagrams were designed on some of the main functionalities in which the system needs.

In figure 4.7 it shows how the event organiser can add an event, assuming the user had registered and logged into the system. The user clicks on the add event button that displays on their account it will then redirect them to the add event page where they could add the name, location date and information of the event which is done concurrently. Once they have entered their details of the even they would click the add event button which would

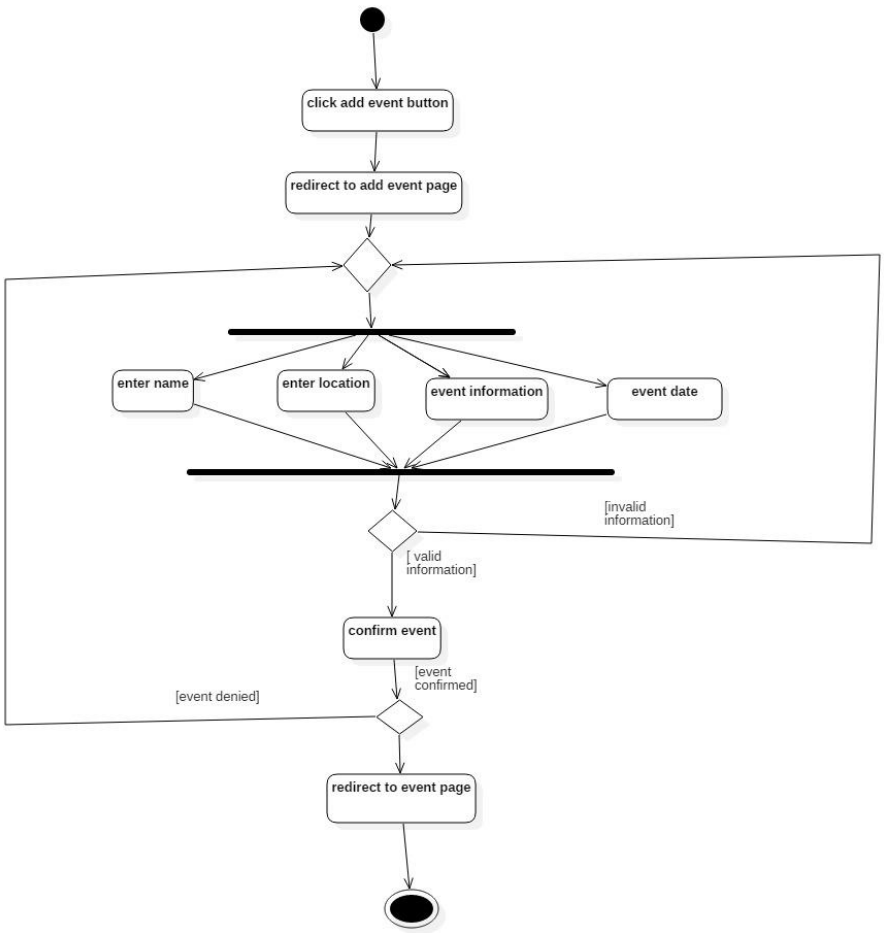


Figure 4.7

confirm the event, however if the details are not valid then they would have to add the details again. If the details are valid then they would be able to view the added event displayed in the view event page.

In figure 4.8 it shows how the event organiser creates a team for an event, assuming that the user has logged in and registered. First the user proceeds to the event page where they would have the option to click on the manage event tab where it would give them option to view all the events that they have created and create a team of volunteers according to the event that they select on. Once they click on the button they can view all the volunteers which are interested in that event. Each volunteer will be displayed followed by their qualifications so that the event organiser knows what role they can be assigned to. Once the event organiser is happy with the selection of volunteers they have the option to confirm the team details. An alternative is that if they are not happy they could always go back change their team details by going back the create team button.

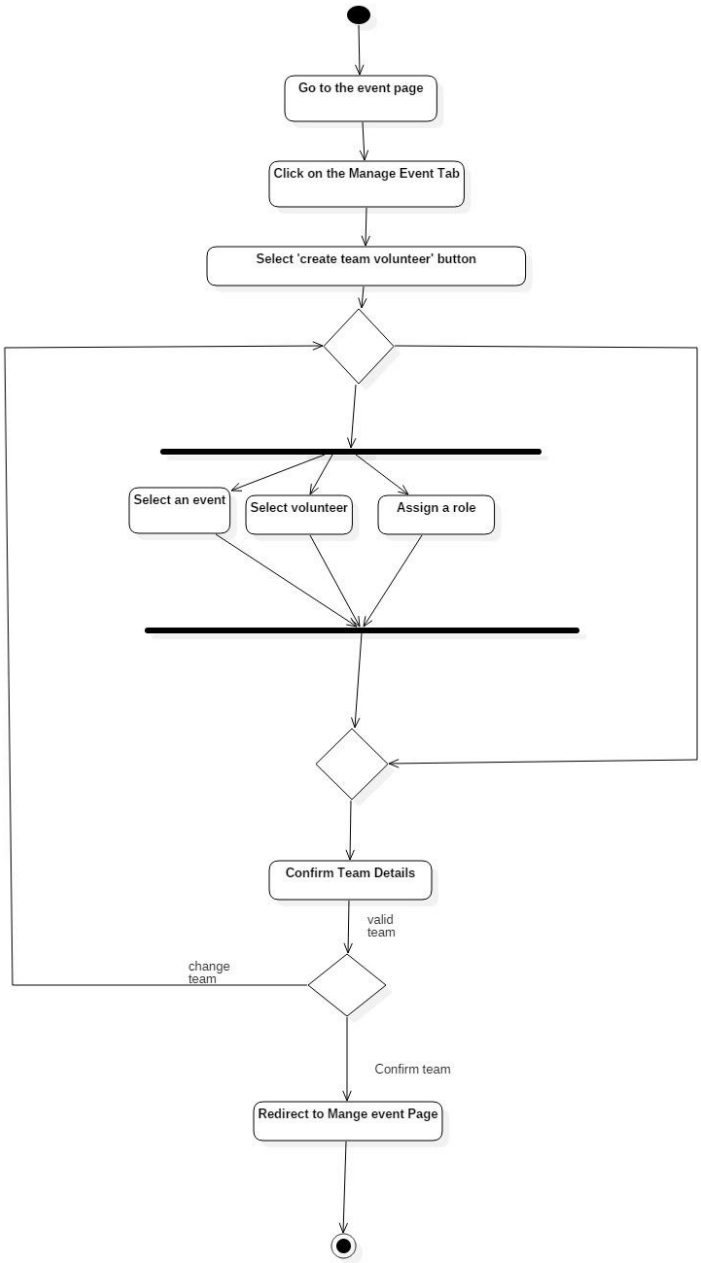


Figure 4.8

4.3.2 User Scenarios:

A user scenario is a written description which describes on how the users might interact with the system by accomplishing a specific task where it can include a cultural information about the user. The reason why user scenarios were used for the design process was because it allows to understand more about how the potential users would use the system rather than thinking about any technological design approaches. In addition, as it is a written description it is very simple to understand the scenario therefore my client wouldn't have a problem in understanding as technological expertise is not needed. Therefore, user scenarios were created for each user type which are going to interact with the system. Furthermore, by creating user scenarios it helps to focus on the user experience side of the application as it shifts the focus from the technical requirements of a task to eliciting emotions with a design. (Armen Ghazarian, 2015)

User Scenario: Scout

Jack is a 12-year-old boy who enjoys attending boat activities every Saturday with his twin sister Emily. He is very adventurous and enjoys being able to have fun and take part in new activities rather than being stuck at home indoors doing homework. He is very keen to take part in events as he just managed to complete his training activity for sailing. Jack has logged in to his account and goes to the event tab in which he is able to view all the upcoming events very easily. He manages to see an event he thoroughly wants to go. He clicks on the event and register his interest. Jack then logs out and goes off and gets ready to go cinema with his best friends from school.

User Scenario: Volunteer

Michael is a 32-year-old sports coach for a secondary school near his home. He enjoys organising and delivering new range of sports activities every week as he believes it very important for people to stay active. He enjoys going shopping and going meeting up with his friends in the park to play football. He also enjoys helping the youth so he takes up his time and commits in helping the weekend activities when he can. Taking his laptop Michael logs into his account and navigates to the event tab where he can view the list of events. As he has a life guard qualification he scrolls down to find an event that is suitable for him. He clicks for a relevant event and realizes he is available to help on the day of the event so he registers to the event. He later receives an email and confirms the event. Since he is already logged in he decides to update his availability for next week as he has relatives coming over to visit him.

User Scenario: Event Organiser

Helen is a 40-year-old woman who has for many years been part of the scout committee and commits most of her time to. As she has been a scout from an early age she has a lot of experience in taking part in events and organising them. Its 12th April deadline for the volunteers and scout's confirmation have passed. Helen decides to create a team of volunteers by check box listing the correct number of volunteers for the day of the event against their qualifications and assigns each volunteer with a key role. After organising the event she begins to view the list of events and decides to delete old events.

4.4 Data Modelling

4.4.1 Entity Relationship Diagram

An entity diagram is a data model which is composed of entities, attributes and relationships. These entities are the object that have data stored in them, the attributes are the characteristics of the entities where each entity must have a unique primary key attribute to be able to uniquely identify a record and lastly there are the relationships between the entities these relationships are defined through cardinality notations. An entity relationship diagram therefore provides an outline of the objects that are needed to be able to design the database. Furthermore, to be able to establish a relationship between the entities the primary key from one table becomes the foreign key of another table, to be able to do this foreign key constraints are added to the tables so it would prevent any damaged linked between the tables. However, before the final ERD was created any many to many relationships between tables had to be resolved first for the databased to be in 3rd normalisation.

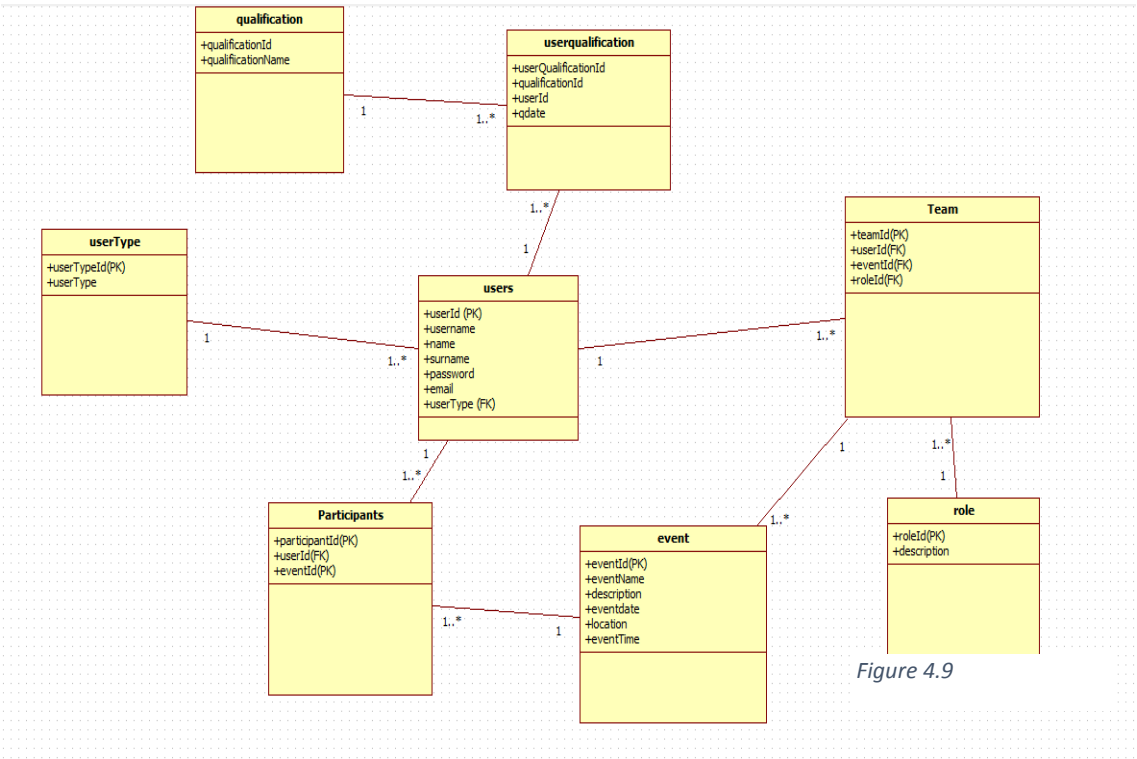
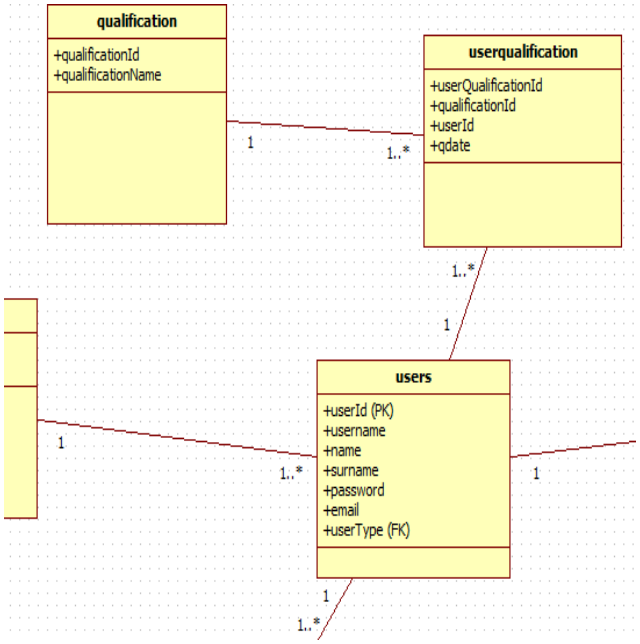


Figure 4.9



In figure 5.0 a many to many relationships is found between qualification and users entity. The reason for the many to many relationships was because a user can have many qualifications and a qualification can belong to more than one user. The relationship was resolved by creating a 'userqualifcation' entity which formed in the primary keys from the qualification table and the users table. The purpose of this new entity was to capture each qualification that the user attains which meant that the date in which the qualification was achieved had to be place in the new entity as each user has different dates in which they qualify for the qualification.

Figure 5.0

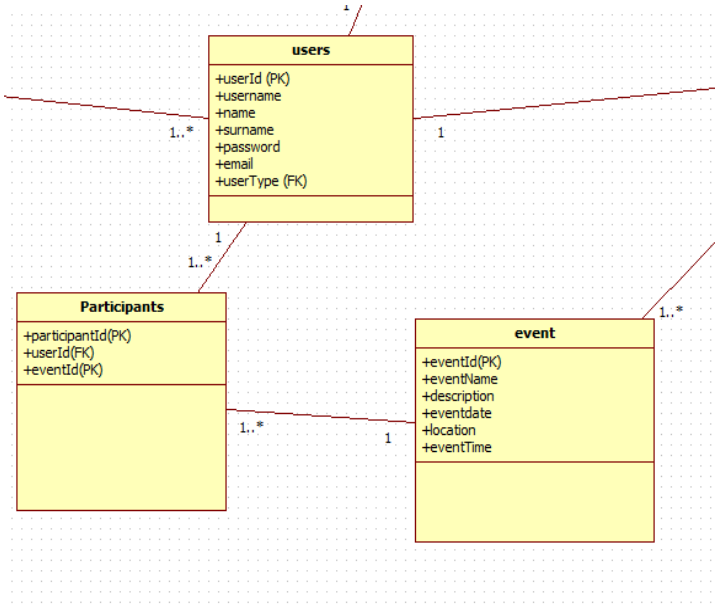
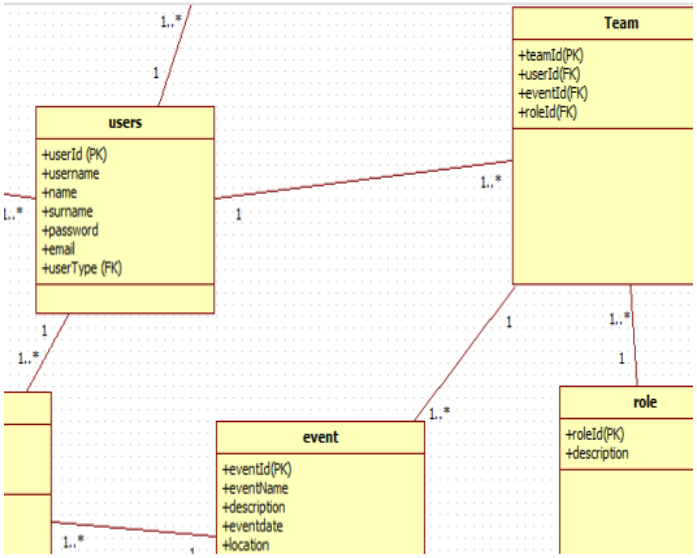


Figure 5.1

In figure 5.1 a many to many relationships is found between users and the event entity. The reason for the many to many relationships was because an event can have many users and a user can take part in more than one event. The relationship was resolved by creating a participant entity which formed in the primary keys from the users table and the event table resulting to be foreign keys. The purpose of this new entity was to make sure that the database captured each event that a user is competing in, in this case it would be the scouts who will take part in the events.

Figure 5.2



In figure 5.2 there is a many to many relationships between users and event entity. The reason for the many to many was because an event can have many volunteers helping and many volunteers can help in more than one event. The relationship was resolved by creating a team entity which formed the user id and event id primary keys being added as foreign keys in this entity. The purpose of this entity was to make sure that it stores a team of volunteers for a specific event and their assigned role. This assigned role comes from the role table where it stores the different roles that each volunteer could potentially be assigned to according to their qualifications.

4.4.2 Data Dictionary

A data dictionary was created to be able to understand the structure of the database. Each of the tables below consist of an attribute name, the data type, the length, a column that explains whether it has a primary key or foreign key, a column whether the attribute is left null or not and other.

Relation Name	Attribute Name	Date Type	length	PK/FK	Not Null	other
users	userId	int	10	pk	Auto increment , not null	
	name	varchar	50		Not null	
	username	varchar	50			unique

	password	varchar	255		Not null	
	Email	varchar	50			
	userType	int	10	fk	Not null	

Table: 4.1

Table 4.2

Relation Name	Attribute Name	Date Type	length	PK/FK	Not Null	other
userType	userTypeId	int	10	pk	Auto increment , not null	
	userType	varchar	50		Not null	

Table 4.3

Relation Name	Attribute Name	Date Type	length	PK/FK	Not Null	other
qualifcation	qualifcationId	int	10	pk	Auto increment , not null	
	qualificationname	varchar	50		Not null	

Table 4.4

Relation Name	Attribute Name	Date Type	length	PK/FK	Not Null	other
userqualification	userqualification	int	10	pk	Auto increment , not null	
	qualificationId	int	10	fk	Not null	

	userId	int	10	fk		
	qdate	date			Not null	

Table 4.5

Relation Name	Attribute Name	Date Type	length	PK/FK	Not Null	other
Participants	participantId	int	10	pk	Auto increment , not null	
	eventId	int	10	fk	Not null	
	userId	int	10	fk		

Table 4.6

Relation Name	Attribute Name	Date Type	length	PK/FK	Not Null	other
event	eventId	int	10	pk	Auto increment , not null	
	eventTime	Date/time			not null	
	location	varchar	50		Not null	
	eventdate	date			Not null	
	eventName	varchar	50		Not null	
	description	varchar	255		Not null	

Table 4.7

Relation Name	Attribute Name	Date Type	length	PK/FK	Not Null	other
team	TeamId	int	10	pk	Auto increment , not null	
	userId	int	10	fk	not null	
	eventId	int	10	fk	Not null	
	roleId	int	10	fk	Not null	

Table 4.8

Relation Name	Attribute Name	Date Type	length	PK/FK	Not Null	other
role	roleId	int	10	pk	Auto increment , not null	
	description	varchar	255		not null	

4 .5 Discussion:

In this chapter the design of the system was focused which is important because there are different types of users in the system which need to interact with the system for it to be functional. This means that the design of the web application should be able to allow the users to perform common tasks easily and generate interest when using it. Firstly, a website structure was created using sketches where this was a simple template used in all the pages which made the webpages look consistent. Once sketches were created, it helped to give an estimate of how many pages were needed when to create wireframes. Each wireframe that was created demonstrated the design consistency and some interaction when going from one page to another. To improve the wireframes, it would have been useful to use a prototyping tool such as sketch software which is a really good way to produce interactive prototypes, however the limitation there was that the software only supports MAC platforms as the developer didn't have access to a MAC.

Once the user interface design was sorted, various activity diagrams were created to show the flow of some functionalities. The diagrams helped to demonstrate also any alternative flows that the user can take if the correct decision isn't made. Furthermore, user scenarios were created to be able to describe on how the users might interact with the system by accomplishing a specific task. A useful way to represent the user scenarios was if personas were created using a Xtensio which is a software that creates templates which captures the

potential users archetypes, image , personality, goals and frustrations if they were to use the system.

Finally, a data dictionary was created according to the erd diagram that was created. The data dictionary shows all the tables and attributes, constraints and data types that each record will attain in the database. However, looking at the resolving number of many to many relationships it is clear to see that the database structure was not that simple to design as thought and several meetings with the client had to occur to be able to get the final database design.

5 Implementation:

5.1 Introduction:

After implementing the database design and interface design this following chapter would help to discuss the following technologies and architecture that was used to help to implement the project. Furthermore, it will then follow by how the database was created and how key functionalities were implemented. By firstly identifying what sort of technologies are going to be used it will enable to choose the right tools for the project as it would help to produce the system efficiently and effectively. The architecture of the design will also be discussed as this would help to outline the structure of the project. Furthermore, the database implementation would include how the tables were created and how the data was generated into the database. Finally, it will discuss how certain functionalities were implemented and as well as any decisions that were made during the process.

5. 2 Tools, Technologies and Resources:

5.2.1 Sublime Text:

The website is intended to be developed using sublime text this was a new tool that was used as primarily notepad ++ was going to be used at the start of the project but it was disregarded due to finding it difficult to be able to highlight opening and closing brackets and other tags. This tool was also chosen because it allowed to change the GUI background colour of the text editor which enabled to read the code better. It also enabled to install plugins to allow faster shortcuts to be made such as splitting the screen which helps to allow to observe one file to another file more easily. An alternative would be notepad ++ as it is a free source code editor however it is only available on Microsoft Windows, therefore it would be a problem if developers worked on different platforms.

5.2.2 PhpMyAdmin:

To host the system, a local server is used which required in downloading XAMPP which is a free and open source platform server. The reason why this was chosen was because it was easy to install and it included packages such as phpMyAdmin, MSQL and apache. This is very useful at it allows to create the tables for the database. An alternative would be SQL developer which provides a supported GUI that can enhance and simplify the database through running sql statements and creating and deleting database objects.

5.2.3 Bootstrap:

Bootstrap was used to make the website responsive, it was a framework that was used which was easy to adapt alongside with CSS. It is useful as it allows to modify the layout of the website and use pre- styled components for example for alerts and customizing the buttons. By using this framework, it allows to create consistency thorough out the site which is very important for the user experience as it eliminates confusion to the user when accessing the site.

5.2.4 XAMPP:

XAMP will be used to establish a local web server on the computer which would be easier to test the back-end code first before transferring all files to the web host since it provides a local database that will allow the developer to connect with PhyMyAdmin and run any SQL queries on it. Furthermore, it is easy to install and simple to use. An alternative would be to host the site onto the university server as students enrolled are eligible for an account on the KUNET. Furthermore, Site Ground can be also used due to having experience in using it before, therefore a domain name would need to purchase.

5.2.4 FileZilla:

FileZilla is a file transfer protocol client which allows to transfer files by uploading and downloading files from a local computer to a web hosting server. Once the code is fully functional and tested it would then be uploaded to the web hosting server other from the local server which will mean it will be available for the end users to see the system and use it.

5.2.4 StarUml:

StarUml is a free software modelling tool. It is very simple to use in creating very precise diagrams and can be viewed with most web browsers. Therefore this tool will be useful to use during the project as it is very quick to create elements and relationships for important diagrams in UML such as an ERD , activity diagrams and sequence diagrams and class diagrams which was used using this tool. An alternative would be to use Microsoft Visio which was used to create the navigational site map structure as there was a simple template to use, however it was not used to create any UML diagrams as there was previous experience on using StarUml. On the contrary for creating wireframes/sketches in the design phase Invision software could be used which enables to transfer sketch designs into a prototyping interactive tool which will help to enable for the client to be able to understand how the user will navigate through the website more clearly.

5.3 Technologies:

5.3.1Html and CSS:

This project is a web based application therefore HTML is used to define the content of each page on the website. However, it can only create static and plain webpages which is not very useful if the website would want to be dynamic. Therefore, CSS is used alongside with Html as it enables to create a more presentable webpage and helps to separate content with design by implementing inside the <style> tags.

5.3.2 Javascript and Jquery:

Javascript was used to manipulate the pages and makes the webpages more dynamic. One of the uses that javascript was used was when validating the data in a registration form. jQuery is a library in javascript which is used to help to make the content and code much easier.

5.3.3 PHP and SQL

PHP is a server-side scripting language that was used to be able to implement the back-end side of the system. It was used in conjunction with MySQL which is a structured query language that is used to create queries to be able to access the tables and interact with the database. Furthermore, it is a popular scripting language used to create web development applications as it enables to create dynamic website.

5.3 System Architecture:

The architecture is split into two sections the client side and the server side. The client side consists of the client-side technologies; html, CSS and javascript in which requests are made. As mentioned the html holds the content data and named elements from which the javascript can manipulate these elements and request for data and the CSS is used alongside with the bootstrap framework and html. An example of a request can be accepting the user input via a variety of control on a form. Whereas on the server-side it listens to the requests and returns the data. As the website has a database that stores the data, it is retrieved by PHP which is the scripting language being used. By using PHP, it can be embedded with html code and interpreted and executed by the web server which generates the web page document. For this reason, PHP is very useful as it can interface very well with MYSQL database through the interaction of SQL queries.

To be able to handle the database with PHP, PDO (PHP Data Objects) approach was used as it is designed to provide a consistent , vendor agnostic interface to relational databases that with a modern object oriented API. Therefore in figure 5.1 a new PDO object was instantiated where it contains the database vendor called mysql, a MySQL server name which since at the start of the project it was called “localhost” but then changed to “kunet.kingston.ac.uk”, a database name called “dbDk1553539", a user name which was called “root” but changed to k1553539 and a database password. All these parameters were stored as a variable to keep the code readable and passed into the new PDO object. This file that was created was stored in a separate PHP file that way if any changes were to be made for example making a connection to a different web hosting server then it was done easily rather than doing it for every PHP file page or html file page that was created in the project. Furthermore, by allowing to create a separate connection file it allows each page that is created to connect to the database by stating require() function at the top of each page before performing any other functions on the page. If it doesn’t make a connection then require() will generate an error and stop the file from running this is shown in figure 5.2.

```
$pdo = new PDO("mysql:host=$servername;dbname=$dbname", $dbusername, $dbpassword);
```

Figure 5.1

```
require('fypconnection.php');
```

Figure 5.2

5.4 Implementing the database:

The ERD diagram created in the design phase was used to be able to create the tables in the database and the data dictionary. As the diagram gave a graphical and representation of the various entities and their attributes and relationships between the entities it helped to have a clear understanding of the data structure and help to minimise data redundancy. By having the ERD diagram normalised properly then it would reduce duplication of data and make the best use of storage. Furthermore, the diagram helped to identify foreign keys which enabled to prevent data redundancy. The database was managed through phpMyAdmin which was provided when installing XAMPP.

5.4.1 Creating the tables:

The way the tables were created was by using data definition language which was used to

```
Create table event
eventID int(5) primary key auto_increment ,
eventName varchar(250) not null,
description varchar(50) not null,
userID (int5) references users(userID),
eventdate date
location  varchar(50) not null
eventtime time))
```

Figure 5.3

manipulate the structure of the database. This included using ddl in sql statements such as using the 'create', 'alter' and 'drop' commands to modify the tables. In figure 5.3 it shows the event table being created which contains the eventID which is automatically increments when a new event is added. The userID is a foreign key to the users this is because it needs to have access to the users who creates the event, which in this case it is done by the user type called event organisers. When the event organiser creates an event, it needs to insert the name of the event, the description of the event, the location of the event, the date and time of the event. However, by adding the constraints to the tables for example the foreign keys it was done separately where the foreign key column was defined as an index in the referring table which in this example it is userID, then by going to the relation view in the database and selecting the foreign key column and the table it usually comes from which in this case it is users. A foreign key constraint name was given and the on delete and on

update menus was changed to restricted because a value cannot be deleted when a row exists in a referencing or a child table that references the value in the referenced table.

```
CREATE TABLE `users` (
`userId` int(5) PrIMARY KEY AUTO_INCREMENT,
 `username` varchar(50) NOT NULL,
 `name` varchar(50) NOT NULL,
`surname` varchar(50) NOT NULL,
`password` varchar(255) NOT NULL,
`email` varchar(50) NOT NULL,
`userType` int(5) NOT NULL , REFERENCES userType(userTypeID)
```

Figure 5.4

In figure 5.4 it shows the users table which consists the attributes that are needed to be able to log in to use the system. The userId is the primary key of this table where it is auto incremented when a new user creates an account. The email of each user is required as it allows the event organiser to send emails to each scout who are interested to take part in the event and volunteers who are interested to help on the day of an event. For this reason, since there are different user types, when a user registers to use the system they would need to specify what sort of user they are, hence why there is an attribute name called 'userType' which is a foreign key in the users table. The userType is stored in a session where it differentiates what type of user has logged into the system.

5.4.2 Populating the database:

To be able to populate enough data into the database an excel spreadsheet file was created, with each column representing each record on the table and a function was created to generate the vast amount of data which was then saved into a csv file. By using an excel file it saves a vast amount of time as phpMYAdmin graphical interface only allows to insert one record at a time which is very time consuming. An alternative other than generating data into an excel sheet is by writing INSERT INTO statements in the SQL query interface to add new records into the table. However, this is very time consuming, especially if there needs a lot of data so that the non-functional requirements can be met. In figure 5.5 it shows an example on how the different users were inserted into the database using excel. Each attribute of each table was entered in the same order as it was displayed in the database the data was then generated using a random generator online to be able to gather as much dummy data as possible. However before populating the database with data a meeting with the client was organised this allowed to know what sort of data is needed to be stored in each table. An example was the name of the types of events being created, each event must be water sport activity as the system is aimed for scouts who take part in water based events. Furthermore, by discussing with the client it gave an insight on what sort of names to name each record in the database and table names.

Edit	Copy	Delete	30	RGPgPXwg8j	Dione	Scott	DkzVGLEqCU	RGPgPXwg8j@hotmail.com	3
Edit	Copy	Delete	31	2lsyHUX2Cp	Alysa	Christa	of0sh4r8iB	2lsyHUX2Cp@hotmail.com	1
Edit	Copy	Delete	32	ImlSjkRuPY	Edmundo	Shantell	3i9ghfXXeK	ImlSjkRuPY@hotmail.com	2
Edit	Copy	Delete	33	ML7gT3O5qD	Kelley ?	Chasidy	QiEoa6qPrS	ML7gT3O5qD@hotmail.com	3
Edit	Copy	Delete	34	KUB5kJFgKx	Machelle ?	Daina	JODZXHrqYw	KUB5kJFgKx@hotmail.com	2
Edit	Copy	Delete	35	wxfllBtHoF	Mercedez ?	Reanna	Vw4neSRLUT	wxfllBtHoF@hotmail.com	2
Edit	Copy	Delete	36	m14j8tU2kq	Elodia	Raymundo	UvjKa3M4cv	m14j8tU2kq@hotmail.com	3
Edit	Copy	Delete	37	hNm7pX1A4y	Andera	Majorie	FF7afTt86u	hNm7pX1A4y@hotmail.com	2
Edit	Copy	Delete	38	JZ15EnlVSe	Trenton	Kamilah	YjVQKlogTy	JZ15EnlVSe@hotmail.com	2

userId	username	name	surname	password	email	usertype
30	RGPgPXwg8j	Dione	Scott	DkzVGLEqCU	RGPgPXwg8j@hotmail.com	2
31	2lsyHUX2Cp	Alysa	Christa	of0sh4r8iB	2lsyHUX2Cp@hotmail.com	1
32	ImlSjkRuPY	Edmundo	Shantell	3i9ghfXXeK	ImlSjkRuPY@hotmail.com	3
33	ML7gT3O5qD	Kelley	Chasidy	QiEoa6qPrS	ML7gT3O5qD@hotmail.com	3
34	KUB5kJFgKx	Machelle	Daina	JODZXHrqYw	KUB5kJFgKx@hotmail.com	2
35	wxflIBtHoF	Mercedez	Reanna	Vw4neSRLUT	wxflIBtHoF@hotmail.com	1
36	m14j8tU2kq	Elodia	Raymundo	UvjKa3M4cv	m14j8tU2kq@hotmail.com	1
37	hNm7pX1A4y	Andera	Majorie	FF7afIt86u	hNm7pX1A4y@hotmail.com	2
38	JZ15EnlVSe	Trenton	Kamilah	YjVQKlogTy	JZ15EnlVSe@hotmail.com	1
39	uqkpkaR8TL	Marilee	Serita	8hT75mKe77	uqkpkaR8TL@hotmail.com	2
40	0NsVmFEHZt	Kristofer	Beata	8iMGlPZZ2M	0NsVmFEHZt@hotmail.com	3
41	0Z37EcFxFn	Brunilda	Katy	4NJoH0Fy4Z	0Z37EcFxFn@hotmail.com	3
42	LsVo44mItC	Jamie	Theola	9vaabC874V	LsVo44mItC@hotmail.com	3
43	5PBnNRFDgz	Iona	Davis	68X1HrxziV	5PBnNRFDgz@hotmail.com	3
44	66FORRNNYJ	Hal	Lucia	uZ6jYCitet	66FORRNNYJ@hotmail.com	1
45	fNDCFnAA5m	Siu	Andre	a1rftBMBmo	fNDCFnAA5m@hotmail.com	3
46	WWar0kDmEv	Eleonora	Patrice	sh6Eh8184p	WWar0kDmEv@hotmail.com	3
47	O8Jf0zFqvL	Ginny	Ulrike	B0cKZUyPmG	O8Jf0zFqvL@hotmail.com	1
48	drCX9u8d7G	Minerva	Bertram	QYd2Pg6Kzn	drCX9u8d7G@hotmail.com	3
49	d5RptpaGSU	Katelynn	Angelena	eIMwOy8tHb	d5RptpaGSU@hotmail.com	2

Figure 5.5

5.4.3 DML Statements: INSERT, UPDATE, DELETE:

One of the key requirements that was identified in the requirements chapter was to be able add an event, update and delete events which is performed by the event organiser. Therefore, for the event organiser to be able to manage their events, data manipulation statements were used this was to be able to manage data within schema objects. For this

Figure 5.6

reason, to be able to add an event the INSERT statement was used to be able to input from a form and append the entered information into the database. Figure 5.6 shows an example of the INSERT INTO statement used when adding an event. The statement begins by using the command followed by the name of the table of where the data is being inserted into which is stored into a variable. The statement is then prepared and the placeholders are binded with the data that is going to be inserted into the statement this is to prevent sql injection then in the execute function an associative array of variables is passed. The column order is kept the same as the value order so that the query won't insert wrong data into the column or cause an error. The userId is a foreign key in the events table, however to be able to get the id of the event organiser a session variable is used called 'userid' which identifies the user who created the event.

```
$sql= "INSERT INTO event (eventName,location,description,userId,eventdate,eventtime) VALUES (:eventName, :location, :description,  :userId, :eventdate, :eventtime )";

$stmt= $pdo->prepare($sql);
$result = $stmt->execute(array(
            ':eventName' => $_POST['eventName'],
            ':location' => $_POST['location'],
            ':description' => $_POST['description'],
             ':userId' => $_SESSION['userid'],
             ':eventdate' => $_POST['eventdate'],
             ':eventtime' => $_POST['eventtime']
        ));
```

The update statement is also used to be able to edit the event made by the event organiser in case a change needs to be made. Therefore, in figure 5.7 it shows the records in the event table can be easily updated by specifying the name of the table that wants to be updated, the column names which is assigned each to a placeholder and the condition where the eventId match the eventId to its appropriate event that wants to be updated.

```
$sql ="UPDATE event SET eventName= :eventName, location= :location, eventdate= :eventdate, description= :description, eventtime = :eventtime WHERE eventId= :eventId";
if(!empty($_POST['eventName']) && !empty($_POST['location']))
{
$stmt= $pdo->prepare($sql);
$stmt->execute(array(
            ':eventName' => $_POST['eventName'],
            ':location' => $_POST['location'],
            ':eventdate'=> $_POST['eventdate'],
            ':description'=> $_POST['description'],
            ':eventtime'=>$_POST['eventtime'],
            ':eventId' => $_POST['eventId']



             ));
```

Figure 5.7

The delete from statement is self-explanatory as it deletes a record specified in a table. In Figure 5.8 it illustrates the statement which deletes an event from which an event organiser has created. The question mark notation used maps the values onto the question mark where then the sql statement is then stored into a variable and called within the prepared function and finally executed.

```
$sql ="DELETE FROM event WHERE eventId= ?";
$stmt= $pdo->prepare($sql);
$delete = $stmt->execute(array($_POST['eventId']));
```

Figure 5.8

In figure 5.9 the use of insert, delete and update was used to be able to manage the events. This was one of the key implementations during the project as without being able to create an event then there wouldn't be a volunteer team to be able to make sure the event is running safely and there wouldn't be a scout team where it consists of the participants.

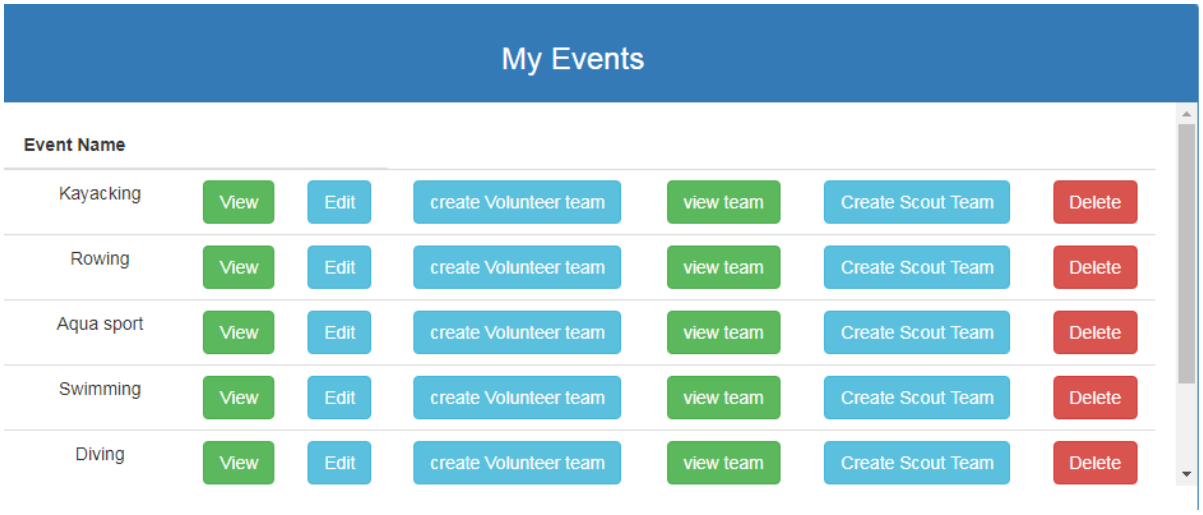


Figure 5.9

Joining tables together:

Several select queries were used to get information and display what was needed. In figure 6 an sql statement was composed of inner joins and aliases to display the event details and volunteers that took part on the day of the event that was created by the event organiser. The aliases were used to be able to retrieve the volunteers name and event organiser name as the data both was accessed from the same table called users. For this reason, an aliase name was given to the column name in the users table which was one for the event organiser and the other volunteer. Inner joins were used to be able to connect tables together so that it returned records that have matching values in both tables which meant if

an event organiser hasn't created an event and assigned any volunteers to the event then there wouldn't be any records being displayed.

```
$sql ="SELECT a.name AS eventOrganiser,
event.eventName,
role.description,
team.eventId,
b.name AS volunteer
from event
 inner join users a on a.userId = event.userId
 inner join team on team.eventId = event.eventId
 inner join users b on team.volunteerId = b.userId
inner join role on team.roleId=role.roleId
 where event.eventId= ?" ;
$stmt= $pdo->prepare($sql);
$stmt->execute(array($_GET['eventID']));
$team = $stmt->fetch(PDO::FETCH_OBJ);
```

Figure 6

5.5 Implementation of key functions:

5.5.3 Login into the system:

One of the main functions to be able to use the system is for the user to be able to log in and access their account. As there are many different types of users by registering and selecting what type of user they are, it will direct them to the appropriate page. It is important to check their user type when they log in as certain user types have access to pages that others do not for example the scouts wouldn't be able to create an event.

For the different users to have access to the system they would need to first register to the website. In figure 12 it shows the user being able to log into the website according to their user type. For example, if their user type is equal to 1 then they would be redirected to the event organiser dashboard, if the user type is equal to 2 they will be redirected to the scout page and if the user type is equal to 3 they would be redirected to the volunteer page. The values one two and three come from the database from the userType table where the types of users were stored separately and not within the same users table.

Figure 5

```php
<?php
session_start();

$message='';

require_once('fypconnection.php');
if(!empty($_POST['username']) && !empty($_POST['password']))
{
  $records=$pdo->prepare('SELECT * FROM users WHERE username= :username');
  $records->bindParam(':username', $_POST['username']);
  $records-> execute();
  $results= $records->fetch(PDO::FETCH_ASSOC);


  if (count($results) > 0 && password_verify($_POST['password'], $results['password']) )
  {
      $_SESSION['usertype']= $results['userType'];
      $_SESSION['firstname']=$results['name'];
      $_SESSION['userid']=$results['userId'];

      if ($results['userType'] == 1)
      {
      header("Location: eventOrganiser.php");
      }
       else if($results['userType'] == 2 )
       {
           header("Location: scout.php");
       }
    else if ($results['userType'] == 3 )
      {
          header("Location: volunteer.php");
      }
 }

    else
    {
      $message='The credentials entered was incorrect, please try again';
    }

}

?>
```

5.5.2 Add an Event

For the event organiser to be able to invite scouts and volunteers to an event, an event must be created. Adding an event consists of the event organiser being able to enter the event details in a form and then clicking on the add event button to be able to submit it to the database this is shown in figure11. All input values in the form must be filled in to be able to create an event as this information is what is being viewed by the scouts and volunteers when they register their interest to a specific event.

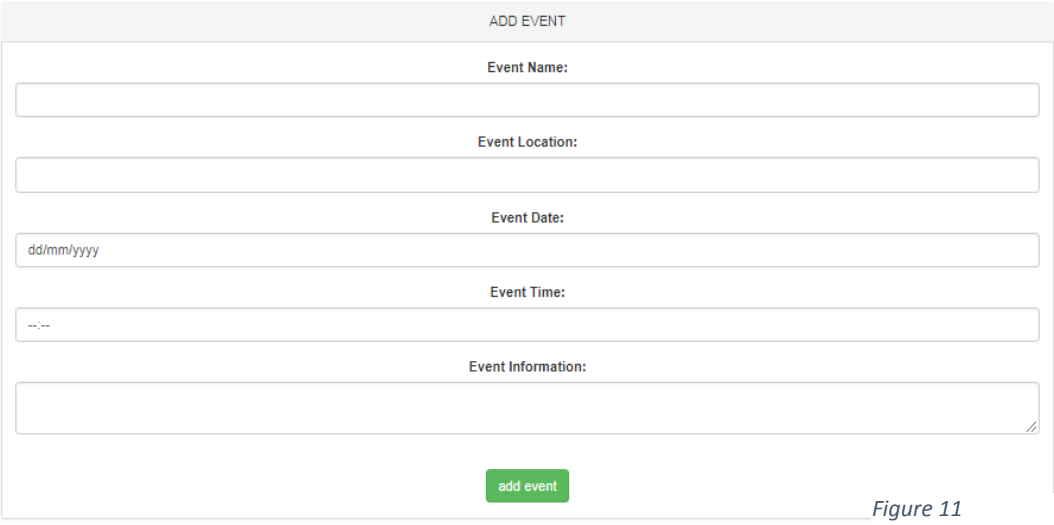


Figure 11

In figure 12 it shows the html code on how the add form was created. The form was created using bootstrap where a panel was used as a bordered box around the form and a bootstrap grid was used to be able to make the form responsive and more dynamic.

```
<div class= "col-md-8 col-lg-8">
  <div id="eventPanel" class="panel panel-default">
    <div class="panel-heading">ADD EVENT</div>
    <div class="panel-body">
      <form id ="event-add" action="insertEvent.php" method="POST">
        <div class="form-group">
          <label for="eventTitle">Event Name:</label>
          <input type="text" class="form-control" name="eventName" maxlength="50" value="">
        </div>

<div class="form-group">
          <label for="eventTitle">Event Location:</label>
          <input type="text" class="form-control" name="location" maxlength="50" value="">
        </div>


<div class="form-group">
          <label for="eventDate">Event Date:</label>
          <input type="date" class="form-control" name="eventdate" value="">
        </div>

        <div class="form-group">
          <label for="eventtime">Event Time:</label>
          <input type="time" class="form-control" name="eventtime" value="">
        </div>


<div class="form-group">
          <label for="eventInfo">Event Information:</label>
          <textarea  class="form-control"  rows"10"  maxlength="1000" name="description"></textarea>
        </div>

        <br>
        <button id="event-add" name="event-add" type ="redirect" class="btn btn-success">add event</button>

      </form>
    </div>
  </div>
</div>
```

Figure 12

5.5.3 Create Volunteer Team:

One of the key implementations during the project was being able to create a team of volunteers for an event. For the event organiser to add volunteers to a team, a list of volunteers would be displayed and according to their qualifications, the event organiser would be able to select what volunteer to be added to the team and assign them to a role on the same page. Therefore, there is a set of checkboxes which are assigned to a variable according to each volunteerId. These variables are run through a for each loop and once submitted it will go to a separate page where the same sql statement would run. In figure 6.1 it shows that a variable name is composed in the form page and then repeated in the update page as this update page doesn't know the variable names, the same sql statement is done again to get their names and rebuild those variable names again. The variable name teamMember is concatenated with its userId and are the users that have been selected and are taken part which is stored in a variable called TeamMember. These variable names

TeamMember and TeamRole are then used to be able to insert into the database which would show the assigned team member and

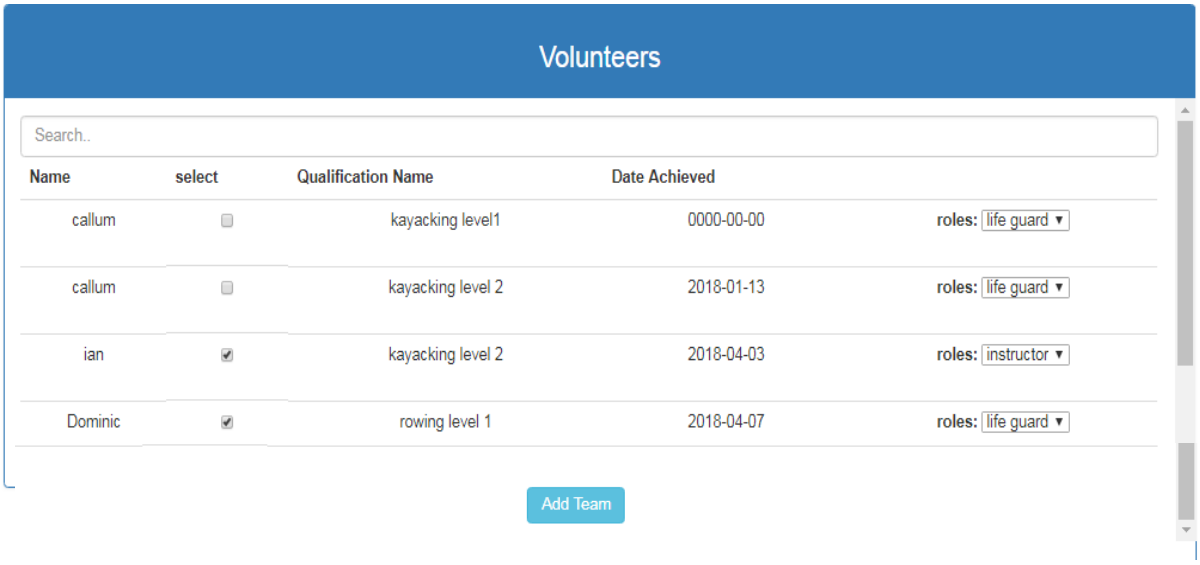
```
 $stmt = $pdo->prepare("
SELECT  users.userId ,users.name, qualification.qualificationName, userqualification.qdate FROM users
 INNER join userqualification
on users.userId = userqualification.userId
INNER JOIN qualification
on qualification.qualificationId=userqualification.qualificationId
" );
  $stmt ->execute();
$vteam=$stmt ->fetchAll(PDO::FETCH_OBJ);

foreach ($vteam as $volunteer) {
  # code...

  $userID = $volunteer->userId; //
  $tempRole = "role".$userID; //
  $tempMember="takingPart".$userID;
  $TeamMember= $_POST[$tempMember];
  $TeamRole = $_POST[$tempRole];
```

Figure 6.1

However, by doing this way it is inefficient as the same sql statement is called twice, therefore an alternative would be to find a way to pass an array of variable names and value which would avoid in creating severable variable names again.



5.5.4 Create Scout Team:

Furthermore, the event organiser must be able to create a team of participants whom are the scouts to take part on the day of the event. Therefore, the event organiser can view the

```
<tr class="record">
    <td> <input name="participants[]" type="checkbox" value = "<?= $scoutID ?>" >
     <td><?=$scout-> name; ?></td>
</tr>
```

list of scouts and is able to select which scouts to take part and submit it. In figure 9 it shows an array was created called Participants which hold more than one value and the value given to each checkbox is the scoutsID.

In figure 10 it shows the array is then looped and for every selected scout that is made it is added into the participant table where the scoutId and eventd is stored. This allows the scout and volunteers to be able to view who is selected for specific event.

```
foreach($_POST['participants'] as $selectedScout){ //index of the list
    $sql= "INSERT INTO participants (scoutId,eventId) VALUES (:scoutId, :eventId)";
$statement= $pdo->prepare($sql);
$result = $statement->execute(array(
            ':scoutId' => $selectedScout,
            ':eventId' => $_POST['eventID'],

        ));
```

In figure 9 it shows the list of scouts which are interested in a specific event. From this the event organiser can select the scouts and add them to a team. Since there is a lot of scouts which are interested in the event and it is first come first basis, the event organiser selects the first numbers of the scouts on the list and adds them to the team by selecting the submit button.

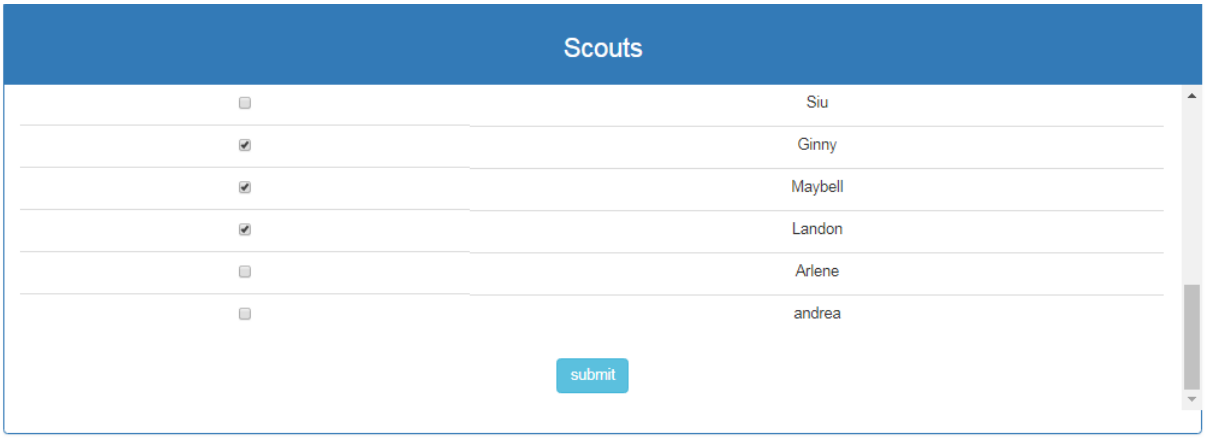


Figure 9

5.5.5 Viewing Teams:

As a scout and a volunteer are put into team they need to be able to know if they have been selected to a specific event of their interest. By being able to view teams it allows to see how many volunteers will be needed at the event and their assigned roles. As each event organiser assigns a volunteer to a role according to their qualifications. Furthermore, by being able to view the team of volunteers and scouts it will also provide information about the event again as a reminder. At first when displaying the team information, the user could click on the view button and it redirected them to a static page where it displayed the information. However, this is not very dynamic to the user to experience when using the system therefore, the information was displayed using a modal dialog. The modal dialog was created using jquery Ajax function. To produce an ajax request from the javascript with jQuery a GET method is used to be able to get the information that is needed in the file followed by the specified URL of the file that contains the information that is wanted to be displayed by the modal. This javascript file is then stored in a variable and then converted into a json object and returned in a JSON representation of the value that is willing to be displayed in the modal. A variable called eventView ID is created using the event.target to be able to get the value of the element of the button when the user clicks on it. This variable is then put inside in the ajax call function and stored as a new variable called teamRequest which is then used to be able to grab and return the selected event as a json object this is showed in figure 1. To be able to grab each element and display it the document.querySelector, a new variable name called myTeam was created to store the data that was passed as a Json object , this made it easier to grab each element this is shown in figure 2.

```php
<?php
//session_start();
require_once('fypconnection.php');


$sql= "SELECT eventName,description, users.userId,users.name,usertype from event inner join users on event.userId=users.userId where event.eventId= ?";
$stmt= $pdo->prepare($sql);
$stmt->execute(array($_GET['teamRequest']));
$team = $stmt->fetch(PDO::FETCH_OBJ);



echo json_encode($team);
```

Figure 1

```
$(".teamViewModal").click(openViewModal);

function openViewModal(event){
  event.preventDefault();
  $("#volunteerTeamModal").modal();
  var eventViewID = event.target.value;
$.ajax ({
  method: 'GET',
  url: "getTeamVolunteer.php",
   data : {
    teamRequest : eventViewID
   },
   success: function (data){
    //console.log(data);
    var myTeam = JSON.parse(data);
    document.querySelector("#eventOrganisers").innerHTML = myTeam.name;
    document.querySelector("#eventNames").innerHTML= myTeam.eventName;
    document.querySelector("#description").innerHTML= myTeam.description;
   }

});
```

Figure 2

In figure 3 it shows the modal dialog being displayed with the event details and the name of the volunteers who are selected to help on the day of the event and their assigned roles. For the modal to stay on the page the event.preventDefault() function was used to not direct the user to a different page when the information was displayed.

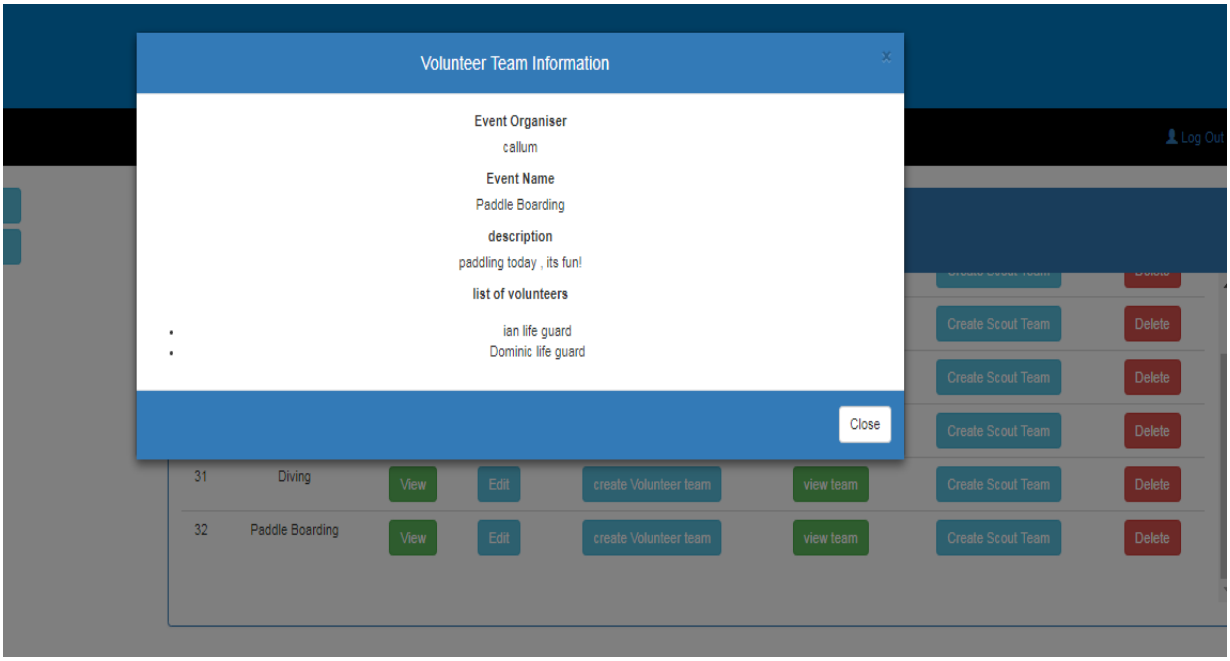


Figure 3

5.6 Discussion

In this chapter the tools and technologies were identified that was used throughout the whole project. The chosen tools were picked because it was believed that it was the most suitable resources for the project. The system architecture was then discussed to be able to understand how these technologies interacted with the client and backend side of the system which was then followed on how the database was implemented. All the requirements that were in the 'Must' section in the MOSCOW was fully implemented and some of the should and could have requirements were also implemented.

Chapter 6 Testing chapter:

6.1 Introduction:

This chapter covers the testing of the system which ensures to point out any defects and errors that were made during the process of the project. It is essential to make sure that testing is done so that it makes sure that there is customer reliability and satisfaction when using the system. The testing will focus on functional, non-functional, validating on database design and usability testing.

6.2 Functional Tests:

6.2.1 Functional test

Below is a list of tests that were carried out using black box testing this was primarily used on the functional requirements which were elicited from the requirements phase. In black box testing the tester only knows the inputs and what the expected outcomes should be and not how the program arrives at those outputs. This benefits the tester because the tester can be non-technical and there is no need for the tester to have detailed functional knowledge of the system.

Function	Procedure	Expected result	Result	What to fix
Add an event	➢ Click events link on the navigation bar ➢ Enter valid details of the event ➢ Click submit	Message saying an event has been added. And then redirecting to the user's homepage	Pass	
Register	➢ Click register ➢ Enter details in each field	User should be alerted to fill out all	pass	

	➢ Click submit	the fields.		
Login accordingly to correct user type	➢ Log in with valid details of event organiser, scout or volunteer	Depending on the user type the user should be directed to their correct dashboard.	pass	
Edit event	➢ Go to manage event tab ➢ Click on edit event button ➢ Make changes accordingly	An alert saying that changes have been made to the event and redirects to the manage event page.	pass	
Delete event	➢ Go to manage event tab ➢ Click delete button to the appropriate event you want to delete ➢ submit	An alert saying that changes have been made to the event and page refreshes.	pass	
Add qualifications	➢ Click qualification tab ➢ Click edit button ➢ Enter valid details of the event ➢ Click submit	An alert saying that a qualification has been added.	pass	
Edit qualifications	➢ Click qualification tab ➢ Click edit button ➢ Enter valid details of the event ➢ Click submit	An alert saying that changes have been made to the qualifications and refreshes page.	pass	
Create scout team for an event	➢ Click scout team button next to the name of the event ➢ Select each member for the team ➢ submit	A list of all scouts that have registered for the event should appear. And an alert saying a team has been created.	pass	
Create volunteer team for an event	➢ Click scout team button next to the name of the event ➢ Select each member for the	A list of all volunteers that have registered for the event should appear. And an alert saying a team has	pass	

	team ➢ submit	been created		

6.2.2 Non-functional Testing:

Below is a list of tests that were carried out using white box testing this was primarily used on the functional requirements which were elicited from the requirements phase. This testing shows the system operated and rather than specific behaviours of the system

Parameter	**Procedure**	Expected result	Result
security	Check the format of the password in the database	The login password must consist of upper case letters and lower-case letters of more than 6 characters	pass
usability	Can view consistent pages.	The system must have a consistent user interface design in all pages,	pass
Efficiency	Check if logging into the system is longer than 5 seconds.	Response time for logging into the system must be less than 5 seconds.	pass
security	Check that the database can take up 50 records.	The database must put up with at least 50 records of data being inserted.	pass

It is important that when logging into the system that each user's password is stored as hashed in the database as it would make it very difficult for an attacker to be able to calculate the passwords easily and apply any brute force attack to the database. Furthermore, the system should be able to have consistent pages as it would help the user be able to focus on the task rather than learning on how to do the task, that way it would increase the users experience. To be able to use the system the users should be able to log in therefore if the user is able to log in no longer than 5 seconds as it would result in delivering a good user experience as the response time would be efficient. As the system is going to be used by volunteers, scouts and event organisers then the database should be able to hold up to more than 50 records, because an event organiser can organise many events and each event will have more than one team of full of volunteers and scout participants.

6.3 Queries:

Before generating a lot of records into the database there needs to be a way in validating the database design. Therefore, SQL statements were generated before creating the database tables as it would be very difficult to drop any tables if there are referential constraints in them such as a foreign key. In query 10, two statements were created to be able to view a team of volunteers, the reason was because at the start a different sql statement was created because to be able to get the name of the event organiser and the name of the volunteer it both accessed the same table called users, therefore aliases was used to be able to get both data. However, when running the sql statement it didn't display the assigned role for each volunteer in an event.

```
$sql ="SELECT a.name AS eventOrganiser,
 event.eventName,
role.description,
 team.eventId,
b.name AS volunteer
from event
 inner join users a on a.userId = event.userId
inner join team on team.eventId = event.eventId
 inner join users b on team.volunteerId = b.userId
inner join role on team.roleId=role.roleId
 where event.eventId= " ;
```

	Queries
1	**Being able to Add an event** $sql= "INSERT INTO event (eventName,location,description,userId,eventdate,eventtime) VALUES (:eventName, :location, :description, :userId, :eventdate, :eventtime);"
2	**Being able to edit an event** $sql ="UPDATE event SET eventName= :eventName, location= :location, eventdate= :eventdate, description= :description, eventtime = :eventtime WHERE eventId= :eventId";

3	**Be able to view event** $sql ="SELECT eventName, description, eventdate FROM event WHERE eventId = ?" ;
4	**To be able to delete an event** $sql ="DELETE FROM event WHERE eventId= ?";
5	**To be able to view qualifications** $sql= "SELECT qualificationId, qualificationName FROM qualification"
6	**To display list of scouts** $stmt = $pdo->prepare("SELECT name, userId from users where usertype='2'");
7	**To be able to sign up** $sql= "INSERT INTO users (username,name,surname,password,email,userType) VALUES (:username, :name, :surname, :password, :email, :userType)";
8	**Add qualifications** $sql= "INSERT INTO userqualification (qualificationId,qdate,userId) VALUES (:qualificationId, :qdate, :userId)";
9	**View qualifications** $sql= "INSERT INTO userqualification (qualificationId,qdate,userId) VALUES (:qualificationId, :qdate, :userId)";
10	**To be able to view a team of volunteers** "SELECT eventName,description, users.userId,users.name,usertype from event inner join users on event.userId=users.userId where event.eventId= ?"; $sql= "SELECT users.name, role.description FROM team inner join users on users.userId=team.volunteerId inner join role on team.roleId=role.roleId WHERE team.eventId= ?";
11	**To be able to list volunteers and their qualifications** SELECT users.userId ,users.name, qualification.qualificationName, userqualification.qdate FROM users

	INNER join userqualification on users.userId = userqualification.userId INNER JOIN qualification on qualification.qualificationId=userqualification.qualificationId ");
12	**To be able to select an event role** $sql= "SELECT roleId, description FROM role"

6.4. Usability Test

To carry out a usability test, 5 people was used to be able to test the system and can navigate around the website. The reason why 5 people was chosen was because according to Nielsen Jacob, "5 people lets you find almost as many usability problems as you would find using more test participants". All participants had general experience in using computers and the internet. To carry out the usability test there was 3 tasks were given to each participant where they had to use the 'think aloud' method which was chosen as it allowed the participants to express their thoughts and allowed myself to observe and analyse their actions on each independent task and encouraged the participant to voice out their opinions if they became too quiet.

Tasks:

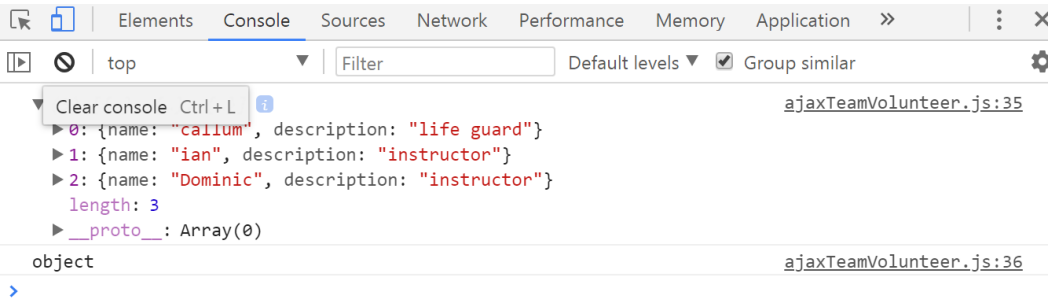
1.	You have an event coming up this weekend, sign up as an event organiser and create an event.
2	Log in as : Joana pw :password You decide that you want to change the time of the diving event to 1:30 pm , once you saved your changes you create your team of volunteers and scout participants for that event.
3	You are interested in the kayaking event that's coming up. Therefore, you update your qualifications and register for the event.

Once the usability test was done was done each participant was handed in a post questionnaire. From the feedback collected all users found the site very easy to navigate through as each core function was under the manage page where it gave a clear indication where the key functions were displayed. However, one of the participants thought it was useful to have links as a back button so that the user can return to the previous page for example when viewing for an event page. They also hoped that instead of directing the pages in a static manner it would be useful to have modal dialogs that way they do not be

directed to a different page. All users found the layout of each page consistent which enabled the users to complete all tasks successfully.

6.5 Console Log:

The chrome developer tool was used to be able to inspect the JSON object. In figure 1 it shows the list of volunteers that are being selected to take part in an event with their assign role. To be able to view this the console.log () function was used to be able to view what the objects contains. This is very useful as it is easy to read and easier to represent null values to know what exactly values are being passed into the model.



6.6 Discussion:

In this chapter it was important to test each stage during the process of the project this included from testing the functional and non-functional requirements in the analysis stage to testing the database design in the design chapter and carry out a usability test to ensure the system was easy to use and whether there were any problems that was encountered when performing the tasks. Some feedback was very helpful such as suggesting limited number of pages when using the system as this would make the system more dynamic. Furthermore, by creating SQL queries and running them in the PhpMyadmin GUI it allows to test whether the right tables are joined together and if data can be retrieved.

Chapter 7 Critical Review:

7.1 Introduction

This chapter will focus on the achievable aspects that came upon when doing the project and the difficulties that occurred during the project that could have been done. Furthermore, any future developments that would improve the project if it was done again.

7.2 Achievements

7.2.1Project areas:

In the analysis phase it has been successful in generating the user requirements as meetings were made at least once a week to be able to discuss and validate the requirements through interviews. Therefore, the chosen methodology was appropriate because client meetings

were arranged and good communication was made between the client and the developer. Furthermore, by using the technique of time boxes it has enabled to control and deliver the functionalities of the project on time.

The design phase has benefited from user experience as the webpages have been consistent throughout the whole website where it allows the user to be able to perform the certain tasks efficiently. The website also benefited from an improvement of the overall usability of the website as particular tasks were triggered by a modal which enables to simplify the overall design of the website, this is very important as the system was created for more than one user type. Furthermore, by going through a design process it has enabled to use different techniques such as producing wireframes which is an iterative design process as it has enabled to allow to communicate with the client and provide early feedback to any changes effectively. In addition, the database was populated with a vast amount of dummy data which has enabled to prove that the system is able to withhold such vast amount of data which replicates to a real-world scenario. Furthermore, the use of bootstrap has enabled to make the website responsive and be utilised in different platforms which is very beneficial for the type of users that are going to use the system. The chosen methodology was suitable for the project as it enabled the developer and the client to have iterative interaction throughout the project as without it the requirements at the start of the project could have gone wrong and the project wouldn't have met its outcome.

7.2.2Persona Reflection:

Before taking on this project a schedule plan was created, from this it has been able to produce a successful project on time as each week a set of deadlines were met. For this reason, none of the project phases were rushed quickly, they were all thought through properly and analysed before making any unrealistic changes. For this reason, if there wasn't a schedule plan created at the start, it would have impacted the system a lot since there was a lot of functionalities that had to be implemented for the system to be functionally used.

This project has enabled to explore different and knew technologies such as using AJAX, where there was no knowledge at all at the start of the project and limited knowledge of PHP, which was vastly used for the back end of the system. However due to the complexity of the project it has enabled to understand more about different languages and has aided to generate complex queries for the database design of the project.

7.3Future Work

7.3.1Project

Certain issues arose when implementing the project as an MVC architectural pattern was not used to implement the code. It would have been beneficial to use this architectural pattern as it would help to separate the code with just the modal, view and controller rather than separating all PHP files and html files. A functional requirement that would have been useful to implement is the email system. By having an email system, it would allow event invitations to be emailed out to all potential interested users. To tackle this problem a code library could be used such as PHP Mailer where it can send emails easily by using PHP from a

web server. Furthermore, this email feature could also include notifications that way scouts and volunteers are aware of an event that has been created by the event organiser. If there was more time to the project the availability functional requirement could have been useful as it would enable the event organiser to select the potential volunteers according to their availability as can be created for the future. To be able to overcome this then an availability schedule would had been implemented for future events.

7.4 Conclusion:

Overall this project has been successful as all the must requirements elicited in the analysis chapter were fully implemented. Furthermore, a fully designed and functional database was also implemented which allowed the system to attain data. A swot analysis was created at an early stage of the project to help to carry out a risk analysis which was later produced to be able to analyse the potential risks that the system may face. In the requirement chapter functional requirements and non-functional requirement were elicited to be able to build a system that matches the end user's needs which was then prioritised. In the design chapter it enabled to visualise how the webpages would be shown to the user this was useful as it allowed the user to have an idea of how the system would look like. In the implementation chapter a database was implemented and any key functional requirements that were needed for the project. Lastly the testing chapter ensured that any functional requirements that were implemented were tested thoroughly so that the when the user uses the system they would not encounter any problems.

Appendix A:

Project Proposal

Abstract:

Many volunteers with different types of skills are needed to run water-based activity events for sea scouts: instructors, safety boat crew, shore support, etc. Each weekend several events might be running each of which needs a team of volunteers. A system is required to store the volunteers and their skills, to request and store their availability over the future weeks, and to create validated teams and send out confirmation requests.

1 Introduction & Background:

1.1 Background

The Scouting association is one of the largest youth organisations in the UK and is the largest national scout organisation in Europe which provides challenging and fun structured activities for boys and girls. One branch of activities that the scouting association holds is the sea scouts which are water based activities for example rowing, sailing, rafting and power boating. However, to be able to do them training should happen. Through training, the sea scouts are taught how to do the activities very safely in a structured way where there are formal qualifications in which they can take. These qualifications are not actually scout qualifications but they are adopted from other organisations for example the RYA association. RYA provide a set of stage one and stage two qualifications for sailing. This allows the training to be conducted safely by completing these courses. Besides that, there is a regular weekly evening session on a Saturday for the scouts in which boating activities are held. However, if it is training activities, invitations are only made to a subset of scouts. This means if a scout has the qualifications already in a certain activity they do not need to attend the training which means that they can attend most events. Furthermore, training doesn't always consist in a classroom, practical lessons are also taught for example knowing how to paddle are one of the key elements when taking part in a water based activity.

Problem

Currently there is no system, as it is heavily paper-based with excel spreadsheets, this proves to be a nightmare for the event organiser. Templates of emails are sent out to volunteers, parents and scouts in which mistakes are made on them. As the predominant system currently used is paper based the problem is a difficulty in managing the spreadsheets because it isn't easily accessible for everyone to see which makes it tougher for information to be obtained and to create a team of volunteers for any event being held. Additionally, many volunteers potentially may have qualifications which meet the desired skills for the event position however all the qualifications are currently kept on one place

which is makes it difficult for the event organiser to maintain and check if the qualification is up to date and keep informed of new qualifications that are being released.

Similarly storing everyone's availability is very hard to maintain because at present doodle surveys are used to ask about people's availability however there isn't any push notifications to say that a volunteer has responded to a survey. So, for this reason lots of paper and excel spreadsheets and this tool to implement are hard to be integrated because only one person can understand the workflow in managing and sending out emails. Due to the current practices, it is impossible to share the task because that one person has a workflow that he/she uses constantly and has a mechanism in processing the data which results in approaching different steps in the workflow. The lack of practice and experience for others to learn causes it very hard to teach others if they can't keep up with the same process therefore there is a need of a new system.

Proposed Solution

The system will include a front end which will be user friendly. This will help users to be able to navigate easily and interact flexibly. There would also be a back end to the system which will only be accessed by the event organiser who will administrate the system. This system will be more streamlined because it will allow events to be viewed by everyone and avoid misinterpreted information being stored. There are great difficulties in finding and organising volunteers to be at events therefore the system will support that through collection of availability and being able to send out invitations. Through this they will be able to report who can come and choose. Furthermore, the system will make sure that there is the right mix of qualifications and be able to see if there is the right number of acceptances so that the event organiser can create a validated volunteer team for the day of the event.

1.2 Stakeholders

There are several stakeholders who will have an influence, power or an interest in the project which can be both internal or external. The stakeholders in this project are:

<u>Volunteer:</u>

The volunteer is the person who is willing to help on the day of the event. Each volunteer will be part of a group of volunteers that will form a team according to their confirmed availability. As the club consists of volunteers the client will be represented by Graeme Jones. The client will be attending weekly meetings to discourse about the project and receive feedback.

Benefits:

The volunteer will be able to receive an invitation to an event efficiently as there would be quicker response time in confirmation. They would also be able to view the event in a responsive format which will benefit from a better user engagement rather than paper based. They will also be able to determine their availability much easier because they would have their own account to be able to update and maintain their availability.

Event organiser:

The event organiser is the person in charge of organising the event and making sure that on the day of the event there is set number of volunteers which are available to attend and help on the day.

Benefits:

The event organiser will be able to create an event and send invitations out to the scouts and parents this will minimise the problem with the current paper based system because presently the event organiser sends out too many errors on the templates of emails. The event organiser will also be able create a team of volunteers which will be very useful as they would be able to check availability and monitor responses.

Scout:

The scouts will be training up for upcoming events. Therefore, they will be on the lookout for any events that are being held at the club so that they can participate and compete. They can be used to test the website as they will have their own dashboard and provide feedback.

Benefits:

The scouts will be able to view the events through their own dashboard this will benefit them as they will be able to confirm their attendance for an event rather than responding to many templates of emails which is one of current problems with the paper based system. As each scout has a certain level of qualification they will be able to update their qualification status and apply for a specific event.

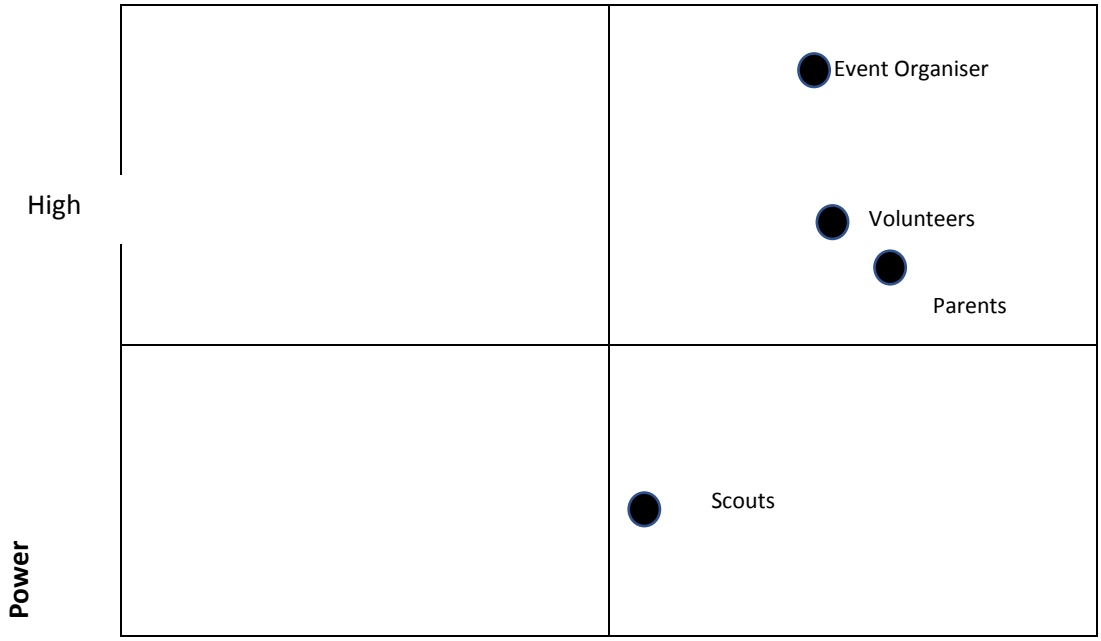
Parent:

The parents approve their scouts' availability for the day of the event. The parents are also volunteers which means that they can be part of the team of volunteers to help on the day of the event, however they do not require a personal qualification to do this.

Benefits:

The parents will be able to view the event and confirm attendance for that specific event through their own dashboard this will be more user friendly for parents as they wouldn't have to navigate around the sites for events. This will minimise their response times in confirming the events which will aid the event organiser in being able to create the event smoothly and improve organisation. This will help tackle one of the problems that is happening with the paper based system.

The figure below shows the power and interest of each stakeholder in the project. By observing the figure above the event organiser is for stakeholders who will have a high interest and high power in creating the events. Therefore, this means that they will hold the most impact on the project success, hence their expectations should be managed closely and engage on a regular basis. The scout stakeholders will be put in the high interest and low power quadrant because they would be monitored and kept informed when events will be running. The volunteer stakeholders will have a high interest and some level of influence because they would be the ones who will be influencing the event organiser in building a team to help on the day of the event therefore it is important to keep them satisfied. The parent stakeholders will too have high interest and some level of influence as they would need to be able to approve their availability as they would be the ones taking their scout to the event.



Low

Low High

Interest

Why is the project important?

To go through with the project a swot analysis has been used to identify the strengths, weakness, opportunities and threats that the project might face externally and internally.

Strengths:	Weaknesses:
• Users will be able to access and view clearly the events. • Users availability will be stored easily through a friendly user interface. • No physical storage as it will be a web application which results in low cost. • Event invitations will be easier to manage. • The system will be able to restrict views depending on the user type.	• Lack of technical knowledge may affect speed of completion. • Web hosting company might get hacked. • Back up storage can get tainted which can result in losing data • Restricted time to complete the system • Other projects involvements may cause delay

Opportunities	Threats
• The system can be expanded for the use of other sport events other than water based events • Promote scouting opportunities for kids. • Increase the number of participants in events and the club becomes more recognised/accredited for a well organised event.	• Revolution in technology which results in the system in becoming out dated • Risk of hackers stealing information about participants.

To conclude by looking at the swot analysis above the point made on the strengths of the project internally outweigh the weaknesses therefore it is safe to continue with the project.

Risk	**Likelihood**	**Impact**	**Mitigating**	**Contingency**
Not getting support from my supervisor.	low	high	Have regular meetings with the supervisor to discuss ideas.	Use other resources example books or internet. Seek help from other supervisors
Poor communication with my client	medium	high	Need to contact with my supervisor to discuss about my project immediately and find alternative ways to contact my client eg skype	Make sure to contact another person within the organisation to be the client.
Losing important documents	medium	high	Need to make sure that regular backup is done weekly and many copies are saved in different	Recover from back-up

			places.	
Lack of technical skill	medium	medium	keep utilising resources online or books.	Seek help from lectures who have the required skills
Database failing	low	High	Hosting company back-up	Recover from back-up

2. Aim & Objectives:

The aim of the project is to promote scouting opportunities for kids by developing an efficient and effective system for organising events.

The objectives:

- Designing a user-friendly website which enables users to access efficiently by April
- Create a database in which volunteer skills and qualifications are stored and populated.
- To develop a system that enables the event organiser to create schedules.
- To comply with DSDM/Atern Agile methodology

3. Technologies & Resources

3.1 Resources

Below is the list of resources that is needed to build the project:

Resources	**Description**	**Alternatives**
FileZilla	to be able to make a connection to the server and can manage files.	
Domain	the unique name that identifies the website	Store everything locally rather than

	on the internet instead of using IP addresses.	
XAMPP	Creates a web server locally.	WAMPP
Notepad ++	Free text editor	Sublime Text 2 could be used as the code is more readable due to the colour coded font however upgrading to sublime Text 3 would endure a cost as a license must be purchased for continuous use. Furthermore, brackets software can also be used.
StarUML	free software modelling tool. It is very simple to use in creating very precise diagrams and can be viewed with most web browsers and it is very quick to create elements and relationships. This tool will be used in	Microsoft Visio can be used to create diagrams as it is more efficient within and between diagrams. On the contrary for creating wireframes/sketches in the design phase Invision software could be used which enables to transfer sketch designs into a prototyping interactive tool which will help to enable for the client to be able to understand how the user will navigate through the website more clearly.
Web Host	Service provider that provides website to be viewed in the internet	The website can be hosted onto the university server as students enrolled are eligible for an account on the KUNET. Furthermore, Site Ground can be also used due to having experience in using it before, therefore a domain name would need to purchase.
PHP MY admin	Manage the Administration of MySQL.	Oracle SQL developer
PHP Cookbook	A book to guide on PHP	

3.2 Technologies

Below is the list of Technologies that is needed to build the project:

Technologies	Description	Skill
Javascript	Object oriented programming language used to make webpages interactive.	Average
Html	it is a front-end language to create website.	High
CSS framework- Bootstrap	Free front-end web framework. Bootstrap will help to make the website very responsive for many different user platforms.	limited
PHP	Server-side scripting language which is used to create an interactive webpage. It also supports databases such as MySQL and Oracle	Average
SQL	It is a structured query language. It is used to create queries to access the tables in the database which can help managing data held in a relational database management system.	High
CSS	It is a style sheet language used to help keep the same format on each web page and it is also used in conjunction with HTML.	High

3.2 Data:

For this project to be developed properly there needs to be sufficient data required for testing queries in the database. Data testing queries is required especially when the database is large because there are complex joins of tables that are going to be used. For example, the event organiser will need to identify all the volunteer's qualifications to be able to create an event role. Therefore, volunteers will need to input their qualifications so that the event organiser can create a team, but before he creates the team, the availability of each volunteer needs to captured. Additionally, the data will need to sensible and within range so that when testing it can help to ensure that every database transaction follow the ACID (atomicity, consistency isolation and durability) properties. The data for implementation will be dummy data, however, my client will be able to give me real data which would help to observe the functionality of the system in the real world for example name of the qualifications that are being held and the type of events.

4 Method & Work Plan:

4.1 Workplan:

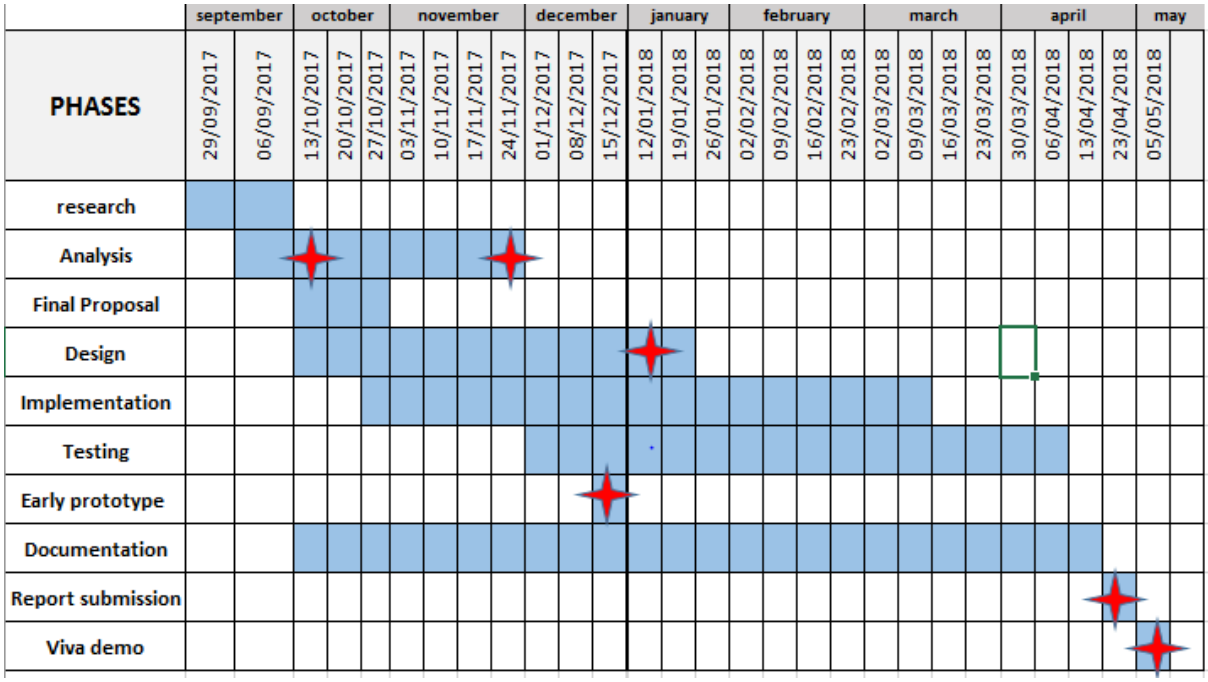
Refer to appendix c which shows the list of weekly deliverables and dates that must be complied with the client to keep the project on track.

4.1 Methodology:

The project methodology and the software approach that will be used to manage the project is DSDM/Atern Agile Methodology. The reason for this methodology is because the goal for using DSDM/Atern is to deliver projects on time and on budget while adjusting for changing requirements along the way. Since there is a fixed scheduled time for the project report submission DSDM/Atern enables to provide an active user involvement always which means that client meetings will be done every week ensuring that the development of the project can be produced iteratively. Additionally, there would be more feedback provided in the DSDM/Atern methodology which demonstrates control and encourages the use of time boxes of about 2-6 weeks which allows us to focus on what is really needed to meet the deadline imposed by the time box.

Extreme programming is not going to be used to manage the software approach because it uses the concept of 'pair programming' and this project is not paired. Furthermore, it uses a lot of daily user involvement which the client and myself would find it very hard to manage and it avoids the overall design and plan of the project. The waterfall approach is not going to be used because it has fixed phases which means once something is done you cannot go back which results in becoming very inflexible therefore the Agile methodology is going to be used because iterations can be done along the way.

Gantt chart:



Milestones:

The diagram above shows a Gannt chart which illustrates the basic flow of the project. It also shows the main phases that will be carried out and milestones which have been indicated with a diamond.

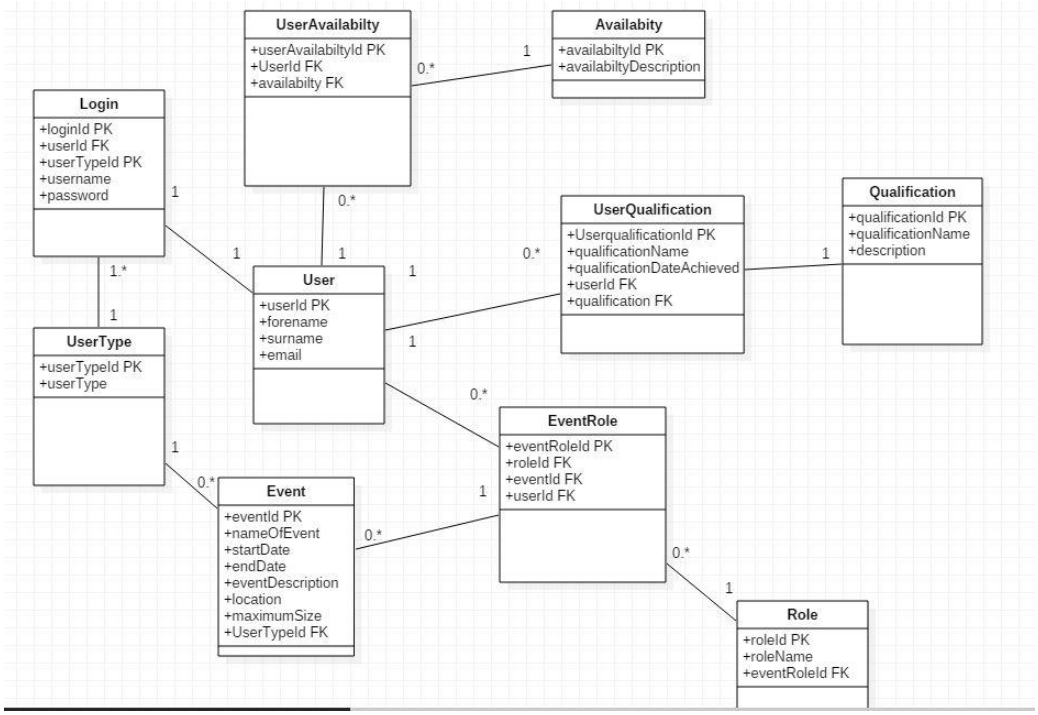
Milestone	Date completed	Description
1	13/10/17	Project Proposal
2	24/11/17	List of requirements/Moscow
3	12/01/18	Revised ERD
4	15/12/18	Early Prototype demonstration
5	23/04/18	Report Submission
6	5/5/18	Viva demo

4.2 Ethics Discussion:

The project will be dealing with a lot sensitive information about everyone including under aged scouts therefore it is important to securely store the data so that the data is only

accessed by those permitted. Therefore, is it important to enforce the data protection act 1998 when keeping data in a database. It is also necessary to obtain parental consent for any individual which is classed as a minor and a fair selection criteria to avoid any selection bias which could relate to the following but not inclusive to age, race, gender, religious views and wealth.

Appendix A:



Initial ERD:

Initial functional user requirements

Parents:

- Receive email
- Confirm email
- Enter availability
- View event
- Login
- Receive information

Volunteers:

- Confirm attendance
- Enter availability

- Log in
- View event
- Determine availability

Event Organiser:

- Check availability
- Create team
- Receive invitation replies
- Invite scouts
- Monitor response
- Create, edit, delete event

Scouts:

- Log in
- View event
- Enter qualifications

Date	*Type*	*Deliverable*
29/09/2017	Documentation	Proposal Appendices
29/09/2017	Methodology	Initial ERD
06/10/2017	Documentation	Complete draft proposal
13/10/2017	Methodology	SWOT
13/10/2017	Documentation	**Project Proposal**
20/10/2017	Implementation	Install DB software
20/10/2017	Methodology	Stakeholder analysis
20/10/2017	Documentation	TOC Requirements Chapter
27/10/2017	Implementation	Establish PHP connection to database
27/10/2017	Implementation	Data and Database Creation
27/10/2017	Methodology	Use Cases and User Stories
03/11/2017	Implementation	Simple SQL and PHP report
03/11/2017	Methodology	List of Requirements / MoSCoW
03/11/2017	Research	Identify Review Topic
10/11/2017	Implementation	Simple SQL insert
10/11/2017	Documentation	Draft Requirements Chapter
10/11/2017	Research	Bootstrap/Mobile API Interface
17/11/2017	Documentation	TOC Design Chapter
17/11/2017	Methodology	Sketches/Wireframes
17/11/2017	Implementation	Inner Join report
24/11/2017	Documentation	Requirements Chapter
24/11/2017	Methodology	Navigation/Sequence diagrams
01/12/2017	Research	Identify SQL queries from requirements
01/12/2017	Implementation	Implement mock-ups of key webpages
01/12/2017	Implementation	Key report 1
08/12/2017	Research	Plan for prototype demonstration
08/12/2017	Implementation	Key form 1
15/12/2017	Presentation	**Early Prototype demonstration**
12/01/2018	Methodology	Revised ERD and Data Dictionary
12/01/2018	Documentation	Draft Design Chapter
12/01/2018	Research	List of key reference material
19/01/2018	Implementation	Key report 2
19/01/2018	Documentation	Design Chapter
19/01/2018	Research	Test Strategy
19/01/2018	Implementation	Key form 2
19/01/2018	Documentation	TOC Review Chapter
26/01/2018	Research	Identify most complex SQL
26/01/2018	Implementation	Key report 3
26/01/2018	Documentation	Draft Review Chapter
02/02/2018	Implementation	Key form 3
09/02/2018	Documentation	TOC Implementation Chapter
09/02/2018		
16/02/2018		
16/02/2018		
23/02/2018		
23/02/2018		
23/02/2018		
02/03/2018		
02/03/2018		
09/03/2018		
09/03/2018		
16/03/2018		
16/03/2018		
23/03/2018		
30/03/2018		
06/04/2018		
13/04/2018		
23/04/2018		

Appendix B

Use Case Descriptions:

Use Case ID:	1
Use Case:	Add event
Description:	Actor wants to add a new event
Actors:	Event organiser
Pre-Conditions	The actor should be logged in
Post Conditions	An event description is added
Basic Flow	4. Once the actor has logged in successfully it goes to its dashboard and presses the 'create event' tab. 5. It leads to a screen where the actor fills in a form which provides details of the event. 6. Selects 'add event' button and displays a success message

Extensions	1a. The actor has entered the wrong username or password which doesn't allow the actor to enter the system 3a. The actor will get a validation message if the event form is not filled in properly.
Performance	Adding the event should be created within 5 seconds
Priority	Must
Trigger	none

Use Case ID:	6
Use Case:	Invites to event
Description:	Actor wants to invite volunteers
Actors:	Event organiser
Pre-Conditions	The event organiser should be logged in and an event is already added.
Post Conditions	Invitation requests have been sent out
Basic Flow	1. Once the user has clicked on a current event it goes and clicks on 'invite volunteer' button. 2. It goes to another screen where there is a table displaying a list of volunteers against their availability and qualifications 3. The Actor check box ticks the volunteers that are available on the event 4. Press the 'submit invitation' button 5. An email is sent out to volunteers 6. The volunteer who receives the email validates the invitation by clicking on an automated link below confirming the event
Extensions	1a. Error message occur if a current event is not clicked. 4a. A message will show up stating no volunteers have been selected to invite.
Performance	Invitation email should be no longer than 30 seconds
Priority	Must
Trigger	This use case is initiated when an event is added.

Use Case ID:	4
Use Case:	view event
Description:	Actor wants to be able to access the event that are available
Actors:	Scouts, volunteers, parents
Pre-Conditions	Actor should be logged in
Post Conditions	Event details are viewed
Basic Flow	1. Once the actor has logged in successfully it goes to its dashboard and presses the 'events' tab. 2. It leads to a screen where the actor can view a list of events that are available
Extensions	none
Performance	Being able to view the event should happen within 2 seconds
Priority	Must
Trigger	none

Appendix D:

Create Tables

```
CREATE TABLE `event` (
  `eventId` int(5) UNSIGNED NOT NULL,
  `eventOrganiserId` int(5) UNSIGNED NOT NULL,
  `eventName` varchar(50) NOT NULL,
  `description` varchar(255) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

```
CREATE TABLE `eventorganiser` (
  `eventOrganiserId` int(5) UNSIGNED NOT NULL,
  `userId` int(5) UNSIGNED NOT NULL,
  `name` varchar(255) NOT NULL,
  `surname` varchar(255) NOT NULL,
  `eventId` int(5) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

```sql
CREATE TABLE `users` (
  `userId` int(5) UNSIGNED NOT NULL,
  `username` varchar(50) NOT NULL,
  `password` varchar(255) NOT NULL,
  `email` varchar(50) NOT NULL,
  `userType` char(1) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

```sql
CREATE TABLE `interest` (
  `interestId` int(5) NOT NULL,
  `eventId` int(5) NOT NULL,
  `userId` int(5) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

```sql
CREATE TABLE `participants` (
  `participantId` int(5) NOT NULL,
  `scoutId` int(5) NOT NULL,
  `eventId` int(5) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

```sql
CREATE TABLE `qualification` (
  `qualificationId` int(5) NOT NULL,
  `qualificationName` varchar(50) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

```sql
CREATE TABLE `role` (
  `roleId` int(5) NOT NULL,
  `description` varchar(255) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

```sql
CREATE TABLE `team` (
  `teamId` int(5) NOT NULL,
  `volunteerId` int(5) NOT NULL,
  `eventId` int(5) NOT NULL,
  `roleId` int(5) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

```
CREATE TABLE `userqualification` (
  `userQualificationId` int(5) NOT NULL,
  `qualificationId` int(5) NOT NULL,
  `userId` int(5) NOT NULL,
  `qdate` date NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

```
CREATE TABLE `usertype` (
  `userTypeId` int(5) NOT NULL,
  `userType` varchar(50) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

Bibliography

Preece rogers, Beyond human interaction https://arl.human.cornell.edu/879Readings/Interaction%20Design%20-%20Beyond%20Human-Computer%20Interaction.pdf page 281 [Last accessed 08/04/18]

Micheal Curdy , Breaking the fidelity barrier: an examination of our current characterization of prototypes and an example of a mixed-fidelity successhttps://dl.acm.org/citation.cfm?id=1124959 [Last accessed 08/04/18] [Last accessed 08/04/18]

Saul GreenBerg ,Sketching user experience

https://books.google.co.uk/books?id=c-RAUXk3gbkC&pg=PA14&lpg=PA14&dq=%E2%80%9CSketches+do+not+have+to+be+pretty,+beautiful,+or+even+immediately+understandable+by+others.+However,+you+should+be+able+to+explain+your+sketches+and+ideas+when+anyone+asks+about+them.%E2%80%9D&source=bl&ots=x8CKuo_Nfb&sig=bg0DYrBNIRZzvcdimysp_FjJMxo&hl=en&sa=X&ved=0ahUKEwiuqbyCwKrZAhXFLFAKHa5MBjYQ6AEIMDAC#v=onepage&q=%E2%80%9CSketches%20do%20not%20have%20to%20be%20pretty%2C%20beautiful%2C%20or%20even%20immediately%20understandable%20by%20others.%20However%2C%20you%20should%20be%20able%20to%20explain%20your%20sketches%20and%20ideas%20when%20anyone%20asks%20about%20them.%E2%80%9D&f=false

[Last accessed 08/04/18]

EMILY GRACE ADISESHIAH(2010)

4 user research methods to help you create reliable and realistic representations of your target users and design accordinglyhttps://www.justinmind.com/blog/user-personas-scenarios-user-stories-and-storyboards-whats-the-difference/ [Last accessed 08/04/18]

Nielsen Jacob (2010), 10 heuristic evaluations https://www.nngroup.com/articles/ten-usability-heuristics/ [Last accessed 08/04/18]

What is DSDM ?(2018) http://www.selectbs.com/process-maturity/what-is-dsdm[Last accessed 08/04/18]

Nielsen Jacob (2010)https://www.nngroup.com/articles/f-shaped-pattern-reading-web-content/[Last accessed 08/04/18]

Ux Planet (2012)https://uxplanet.org/dark-side-of-ui-benefits-of-dark-background-12f560bf7165 Last accessed 08/04/18]

Armen ghazarian (2015) User stories https://designmodo.com/user-stories-ux/ Last accessed 08/04/18]